How to feed Machine Learning algorithms with proper metrics?

SC2.28

Kai Hartmann & Annette Rudolph

e-learning
SOGA

https://www.geo.fu-berlin.de/en/v/soga/

github

https://github.com/soga-lab/EGU2026_SC

TECHNISCHE UNIVERSITÄT BERLIN

Stiftung Innovation in der Hochschullehre

EGU26 GENERAL ASSEMBLY

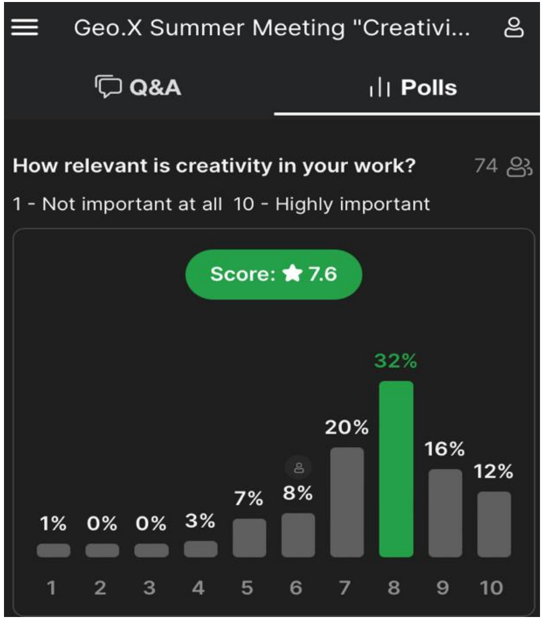
FREIE UNIVERSITÄT BERLIN

Outline

Slides with such blue background are for deeper insights or details
➔ not part of the oral!

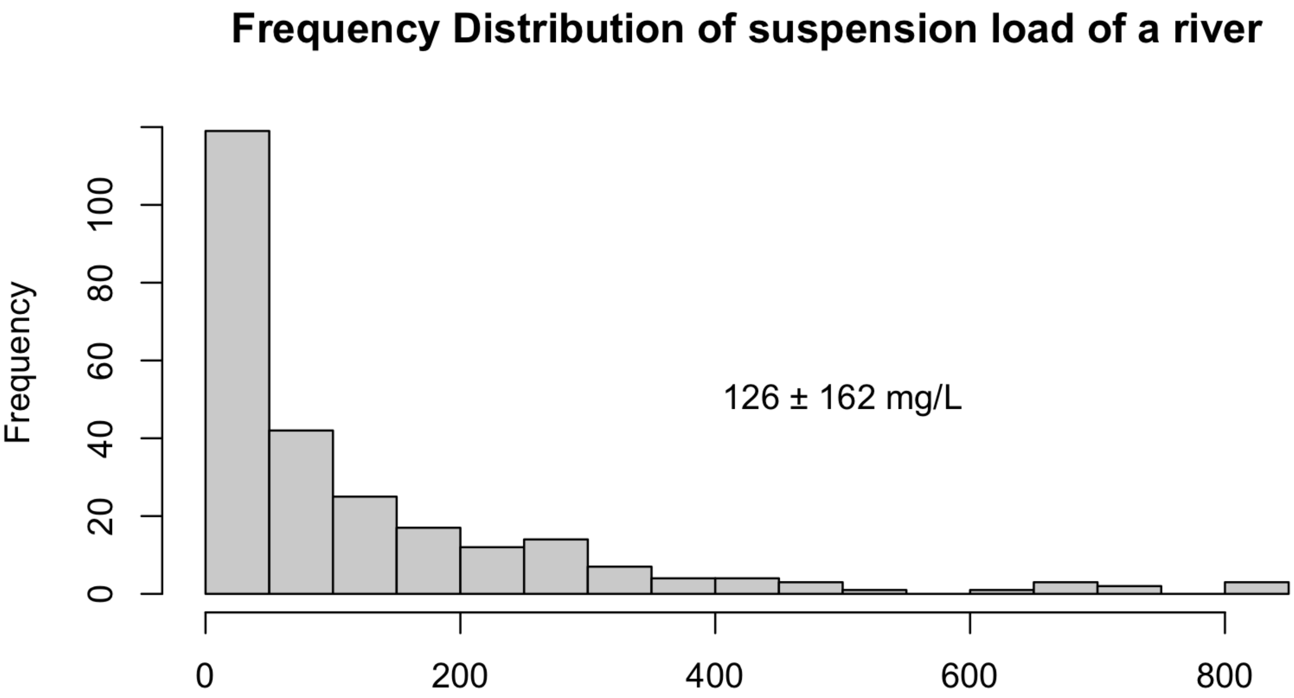
0. Motivation
1. Digesting Measures: How does MLA decide?
2. Meaningful measures: Truth beyond stats textbooks
3. Recipes: Preparing real-world data for ML
4. Hands-on exercises

Once upon a time on a Geoscientific community meeting:



The suggested centre is part of the minority!!!!

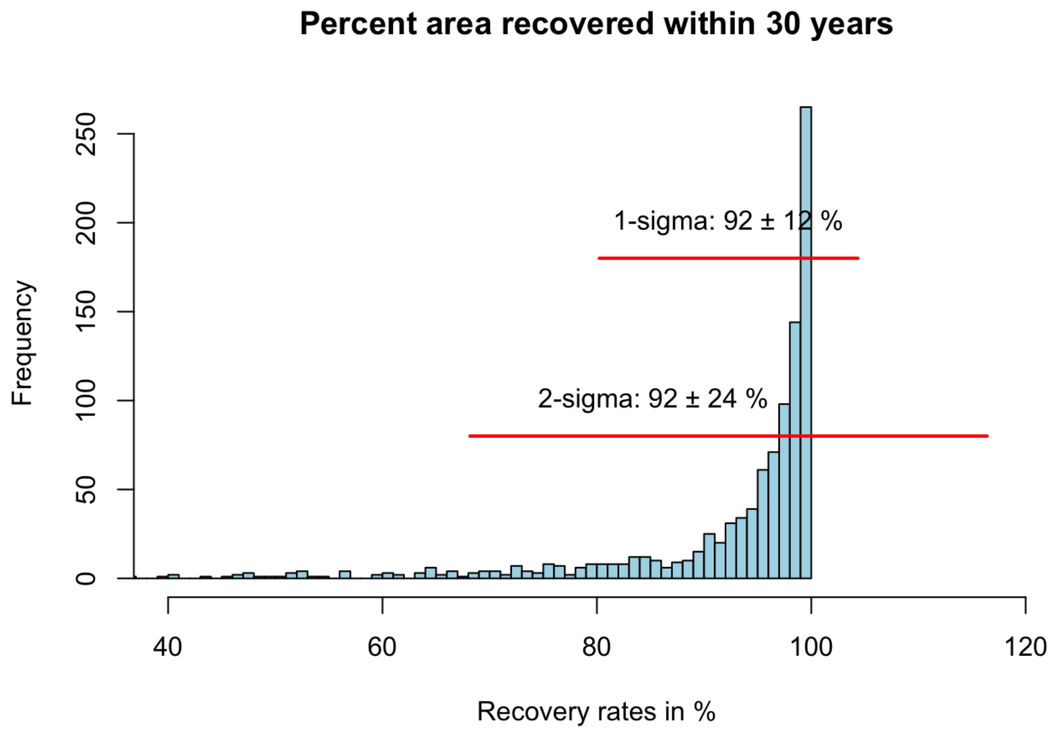
Meaningless measures

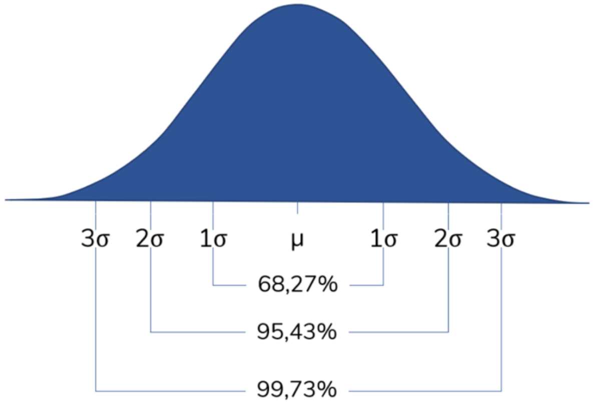


126-162 = **- 36 mg/L**

➔ negative concentration?

Meaningless measures





- Are these very special samples with unfavorable measurements?
- Just an exception?

Meaningless measures

	n = 19					
	Range			Mean ± sd		
	2017	2018	2019	2017	2018	2019
	Dry	Wet	Wet	Dry	Wet	Wet
Temp	20.1–31.2	23.3–33.2	22.4–32.1	25.0 ± 3.3	28.8 ± 2.2	27.2 ± 2.8
pH	7.3–9.4	6.3–8.6	7.0–9.2	8.2 ± 0.6	7.4 ± 0.8	8.1 ± 0.7
EC	80–2230	90–2020	76–2935	911.2 ± 669.9	614.7 ± 655.4	724.9 ± 229.3
Turbidity	17.5–1471.6	47.9–1422.3	16.1–1615.4	631.5 ± 551.6	722.2 ± 545.9	477.6 ± 496.7
Cl^-	3.9–554.0	9.28–678.0	3.9–342.0	206.9 ± 179.0	170 ± 209.1	71.4 ± 105.4
SO_4^{2-}	72.5–707.1	48.7–1224.0	47.0–309.2	252 ± 171.1	298 ± 334.3	126.1 ± 78.9
NO_3^-	0.1–3.5	1.6–24.1	0.2–3.8	1.6 ± 1.1	8.5 ± 6.9	1.0 ± 1.1

- “I reported my data in this way, **because everybody is doing this.**
- “I want to keep my data in an **comparable way**”!
- “This just happened because of **the large variance!**”
- “The **sample size was to small** for a more accurate exstimation!”
- “Don’t take this too serious, it is **scientific data reporting standard**!”
- “I just wanted to **show the mean and the variability** of my data!”
- “I’ve tried the statistical right way, but then **my story has disappeared**!”

Did you heard another excuse? Please share it with us in the comments!

Outline

0. Motivation

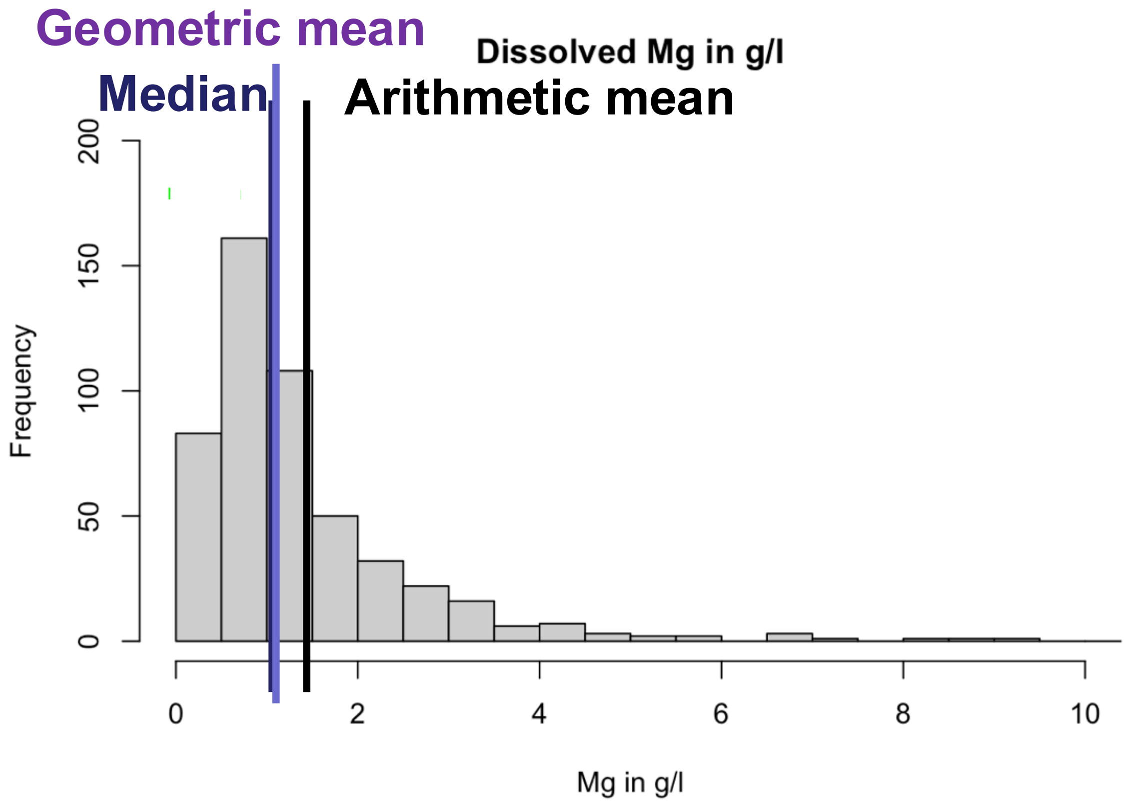
1. Digesting Measures: How does MLA decide?

2. Meaningful Measures: Truth beyond stats textbooks

3. Recipes: Preparing real-world data for ML

4. Hands-on exercises

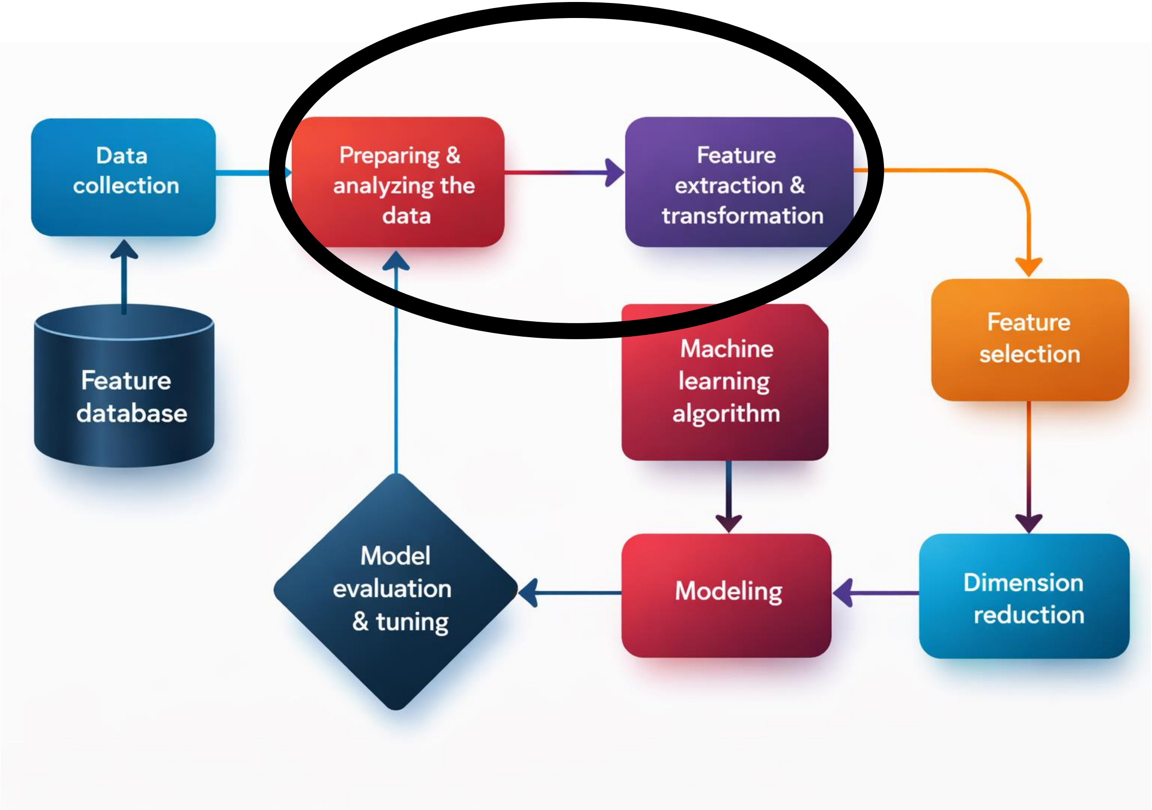
Why do we need proper metrics?



- **Median 1.05**
- **Geometric mean 1.04**
- **Arithmetic mean 1.42**

→ $\mu - 2\sigma < 0$

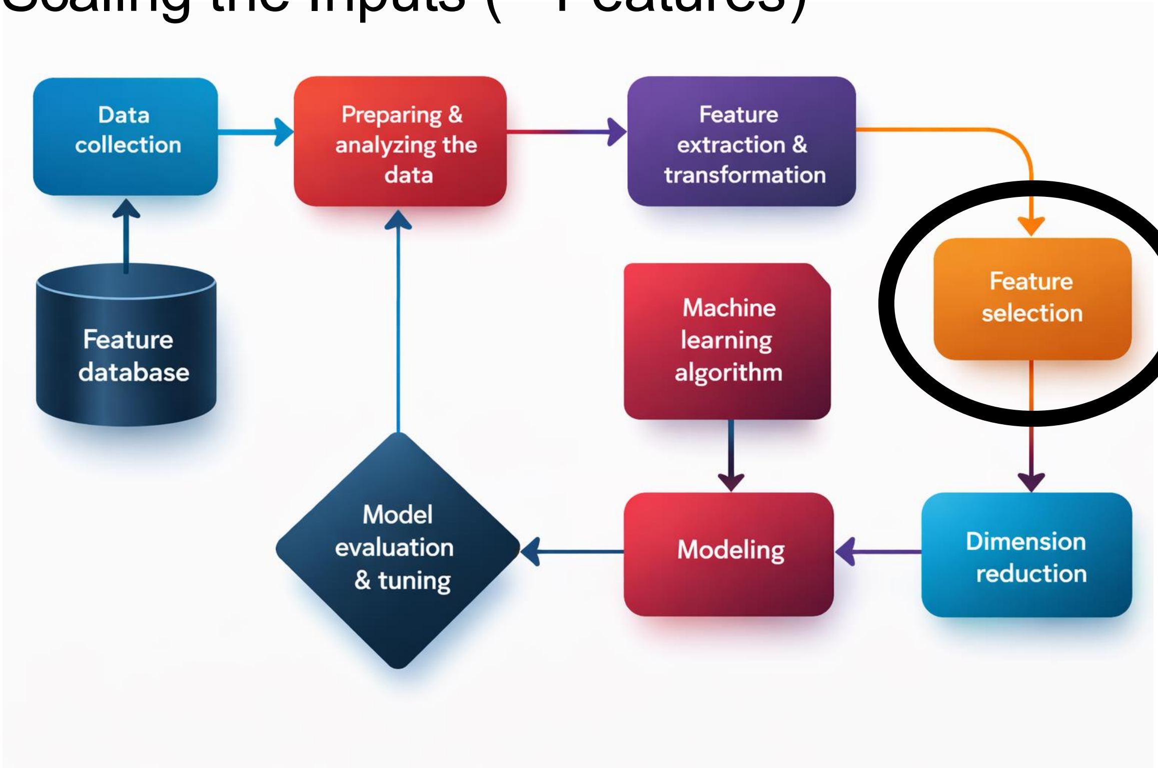
How does MLA decide?



Why is data transformation crucial?

How does MLA decide?

1. Choosing/Scaling the Inputs (= Features)



Based on correlations

Why do we need proper metrics?

- Given paired data points of *n* pairs $\{(x_1, y_1), \ldots, (x_n, y_n)\}$
- The Pearson Correlation Coefficient is defined as:

Arithm. Mean

Covariance

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} = \frac{s_{xy}}{s_x s_y},$$

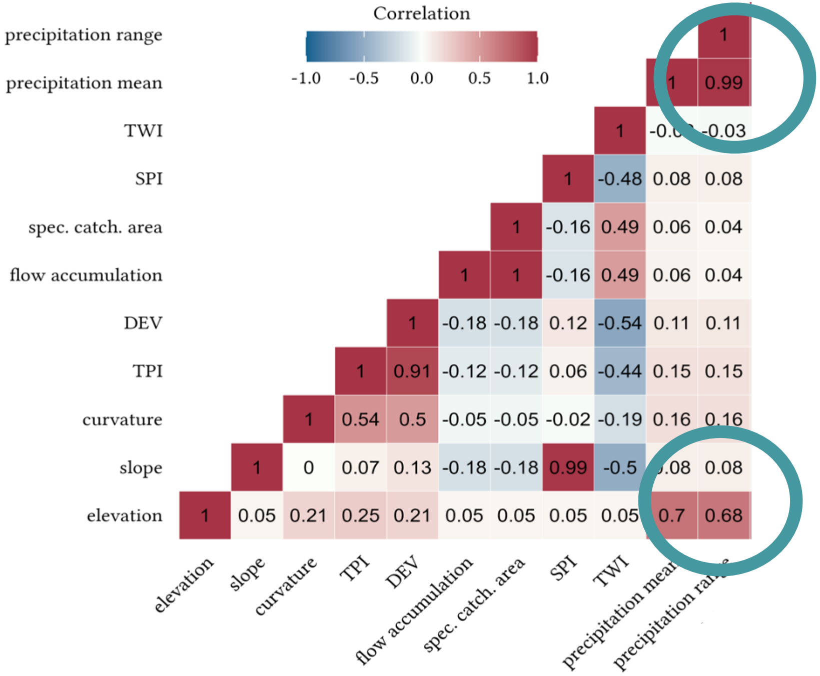
Standard Deviations of x and y

- → Spurious correlations without FST

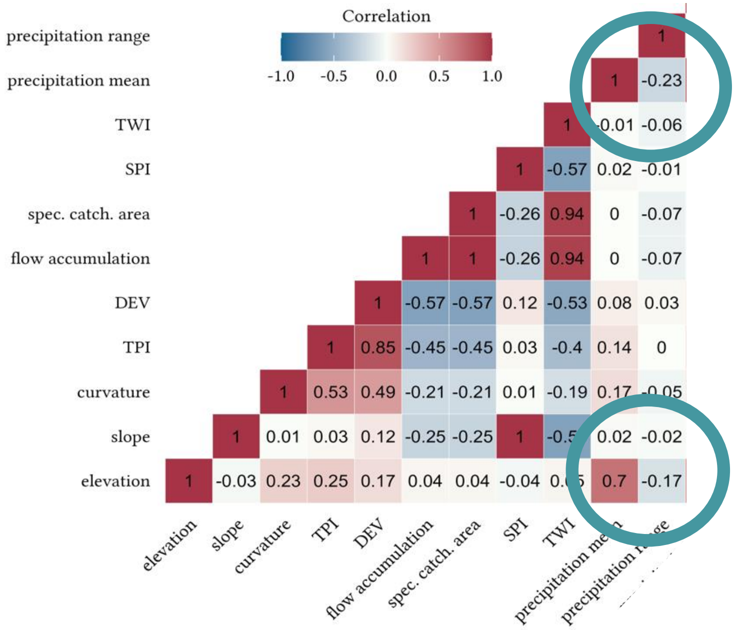
Why do we need proper metrics?

- Example: Gully Susceptibility in Lesotho (Frenzel et al, in prep.)

Before FST

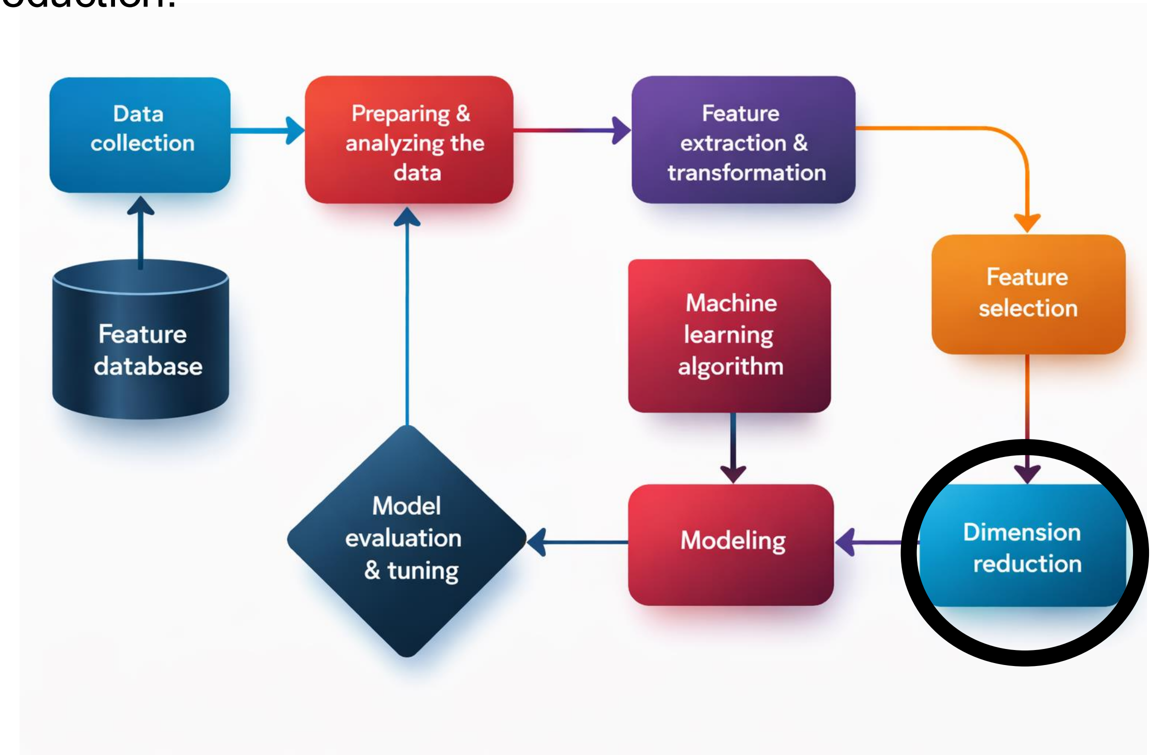


After FST



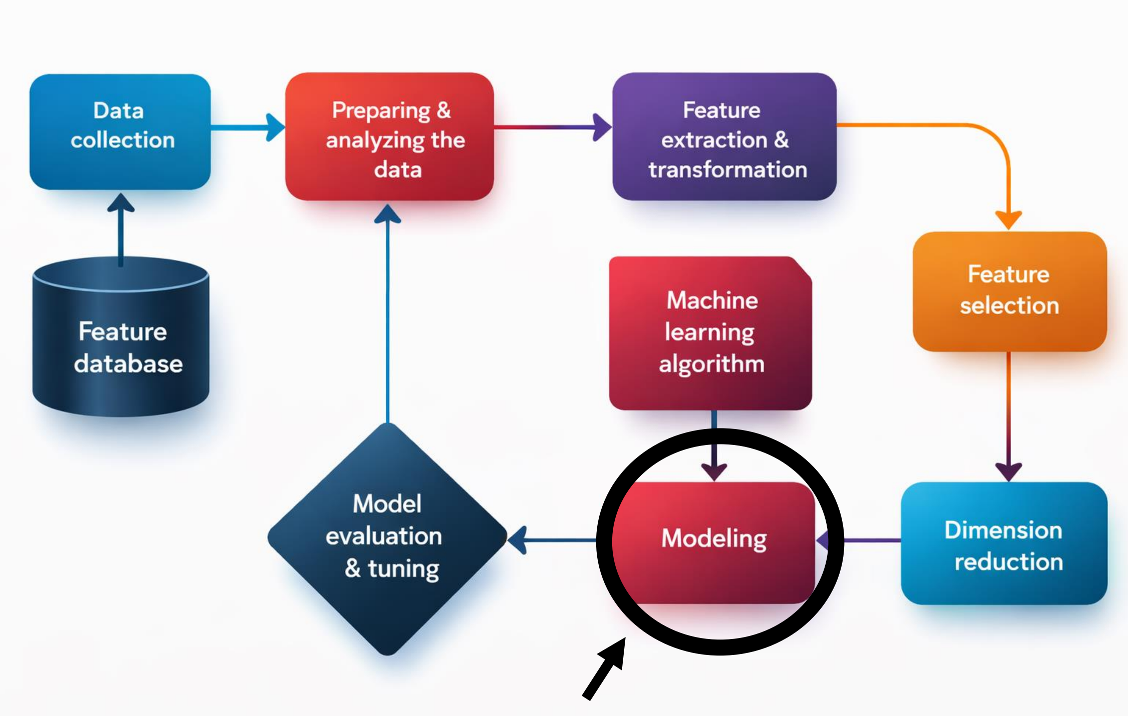
Why do we need proper metrics?

- Dimension reduction:



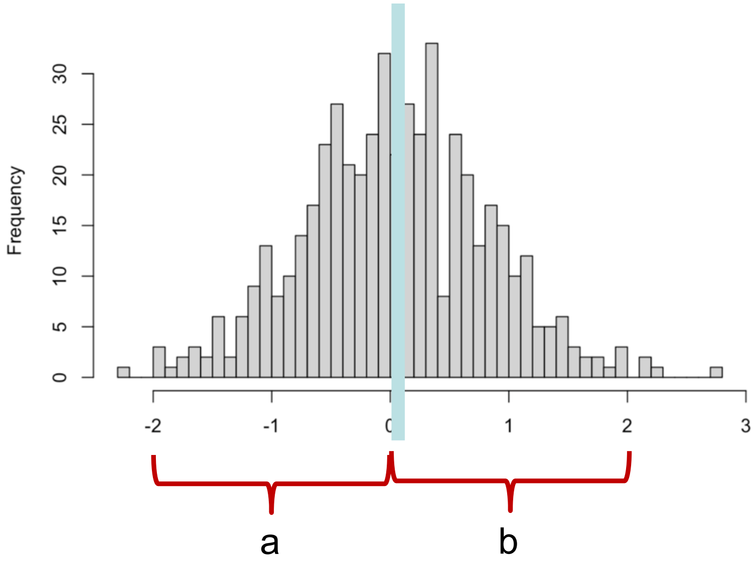
PCA:
calculation
of covariance,
variance
depend
on the mean

Why do we need proper metrics?

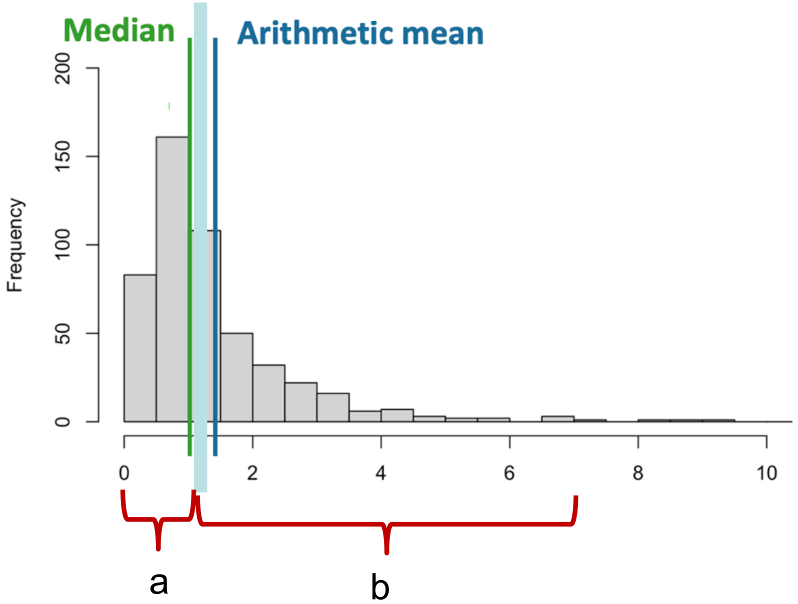


A lot of algorithms are **distance based**, e.g. KNN, SVM, k-means, …
→ We need **correct metrics**!

But…



Symmetric distributions:
Euclidian distances make sense
a = b



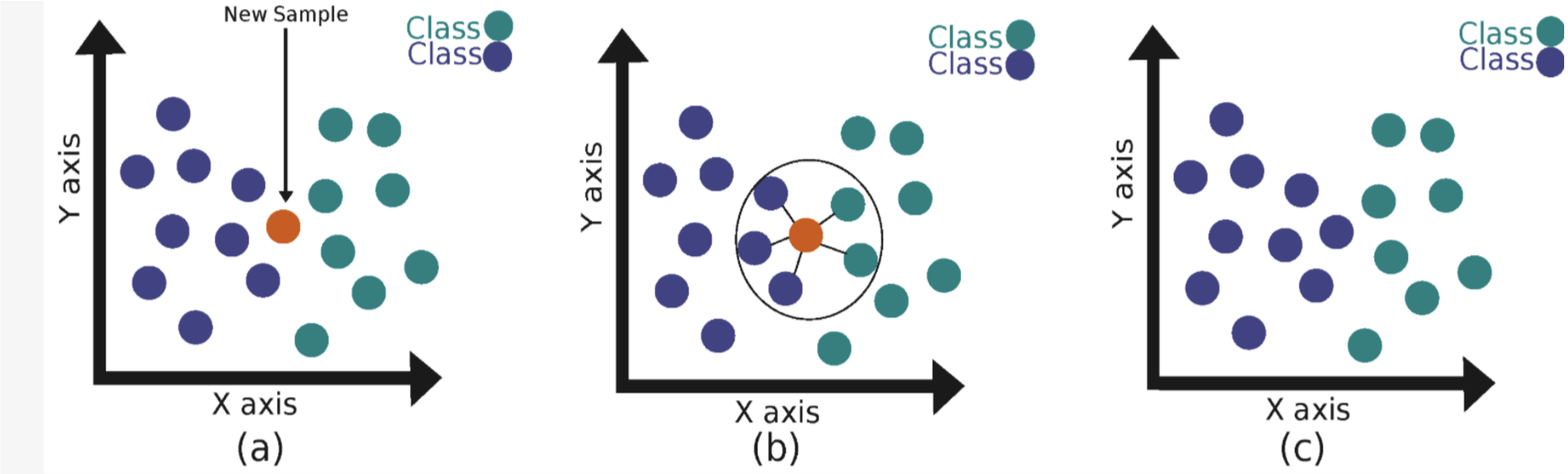
Non-symmetric distributions:
Euclidian distances do not make sense
a ≠ b

Distance based algorithm example I: KNN

K-NN with K=5
Orange: New Sample / Unknown instance

Find k nearest neighbors

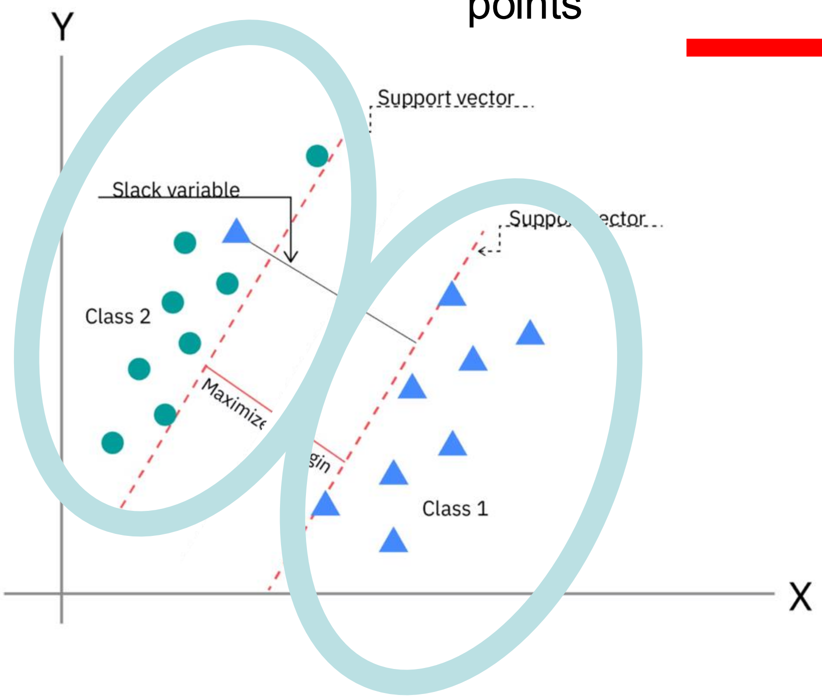
Which class predominates?

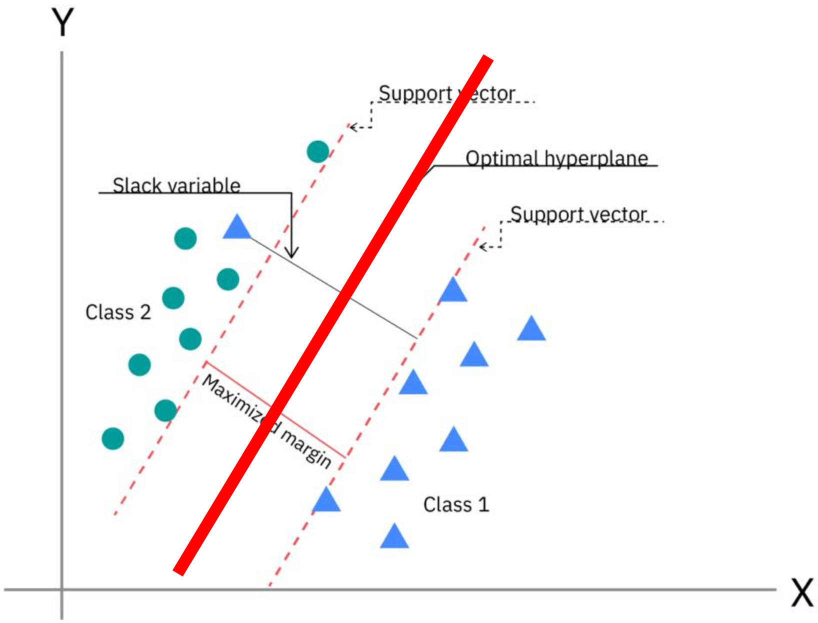


Maldonado-Romo, A., Montiel-Pérez, J. Y., Onofre, V., Maldonado-Romo , J., & Sossa-Azuela , J. H. (2024). Quantum K-Nearest Neighbors: Utilizing QRAM and SWAP-Test Techniques for Enhanced Performance. *Mathematics*, *12*(12), 1872. https://doi.org/10.3390/math12121872

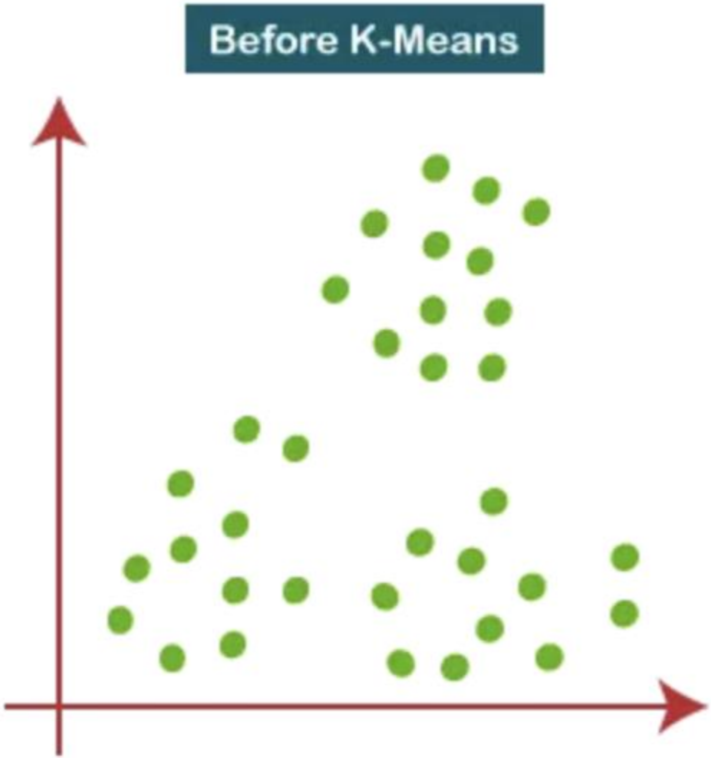
Distance based algorithm example II: SVM

Find the line separating the data set that has the **maximal distance** to the data points





Distance based algorithm example III: k-means

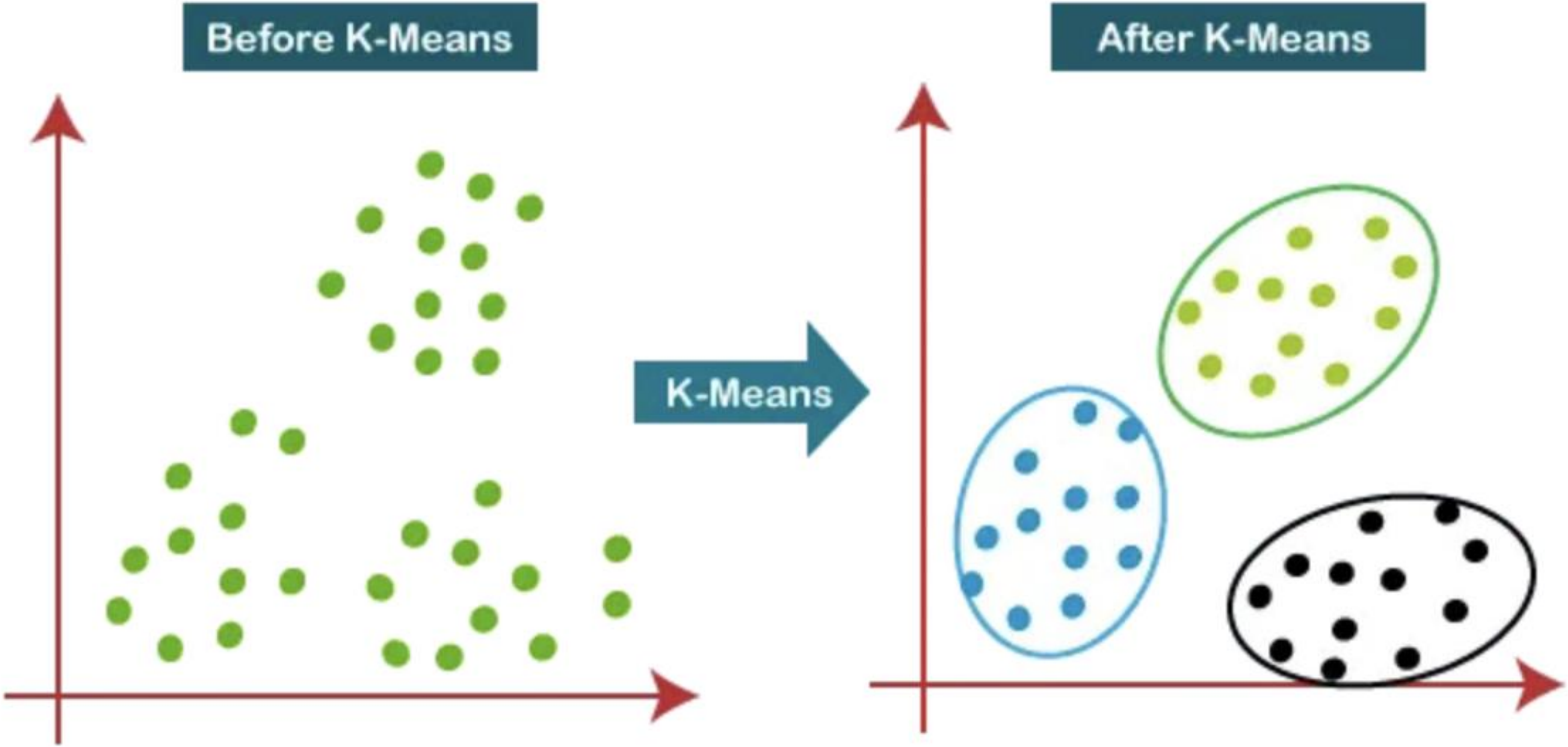


1.Specify K and Initialize: Choose the number of clusters *K* and randomly select K*K* initial cluster centers
Assign Points to Clusters: For each data point, **calculate its distance** to all centroids and assign it to the nearest one, forming temporary clusters.

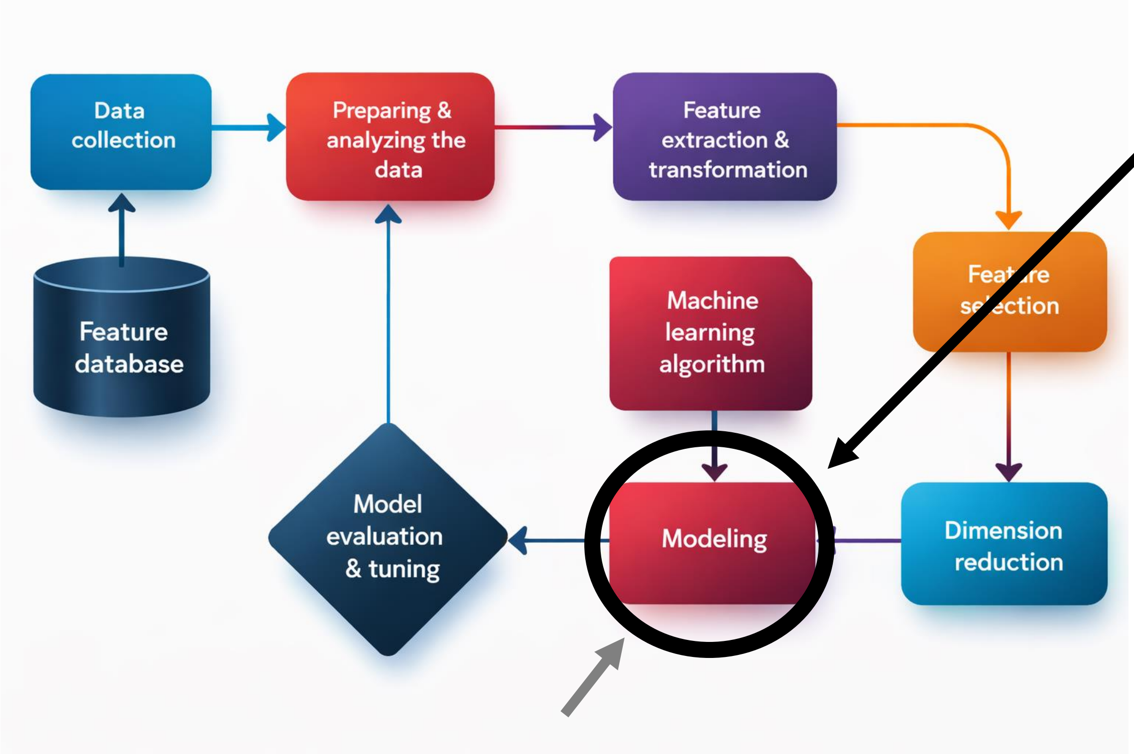
2.Update Centroids: Recalculate each cluster center as the **mean (average)** of all points assigned to its cluster.

3.Repeat or Converge: Iterate steps 2 and 3 until centroids stabilize (change less than a threshold), assignments no longer change, or a maximum iteration limit is reached

Distance based algorithm example III: k-means



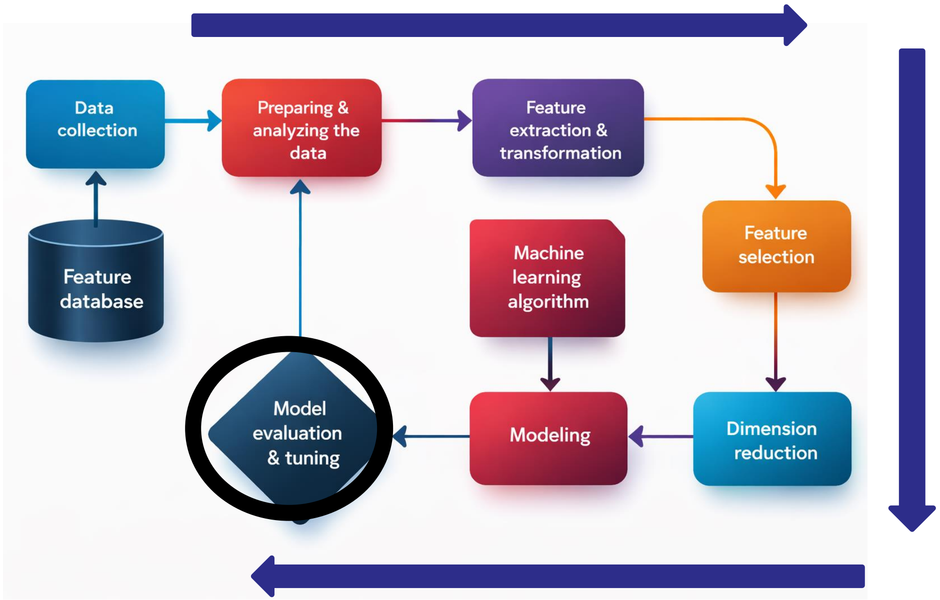
Why do we need proper metrics?



- But also…
  → **Regression (based)** algorithms need normality (e.g. NN)
  → **Gradient** based algorithms
  → **proper coordinate systems are needed!**

A lot of algorithms are **distance based**, e.g. KNN, SVM, k-means, …
→ We need **correct metrics**!

Why do we need proper metrics



Neglecting Feature Scale Transformations (FST)s lead to unreliable And lower scores

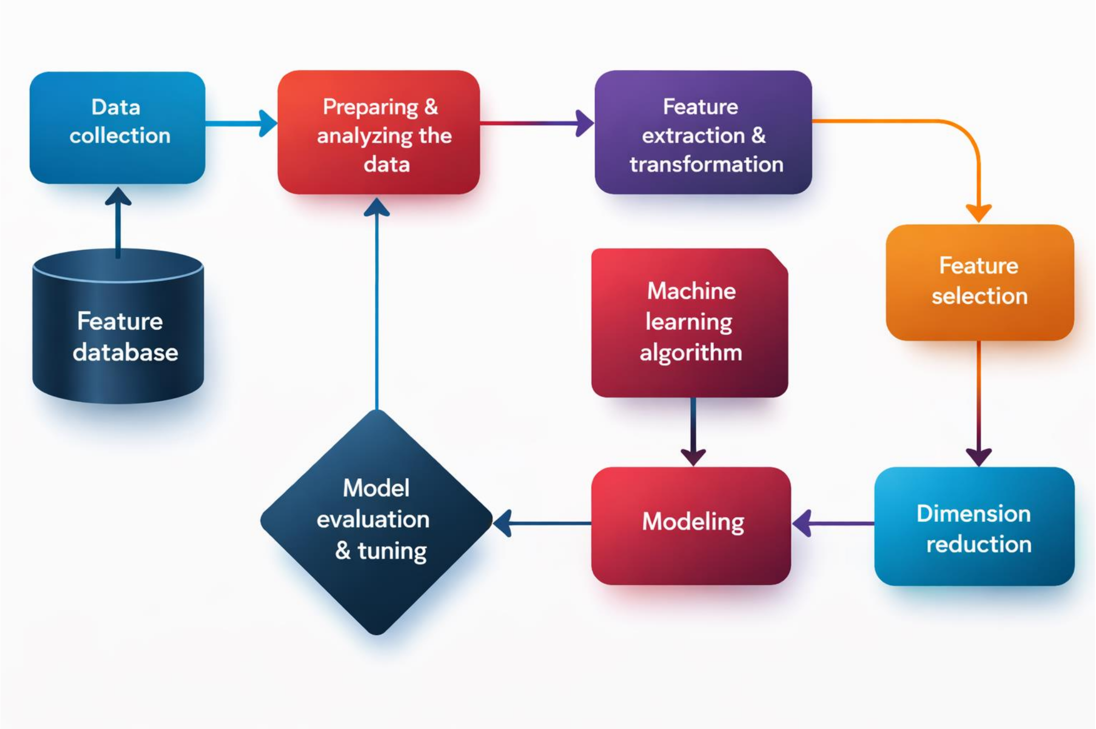
- Wrongly applied models (e.g. that assume normal distributions and/or are based on metrics

  → Non-normally distributed residuals

  →Accuracy scores become meaningless

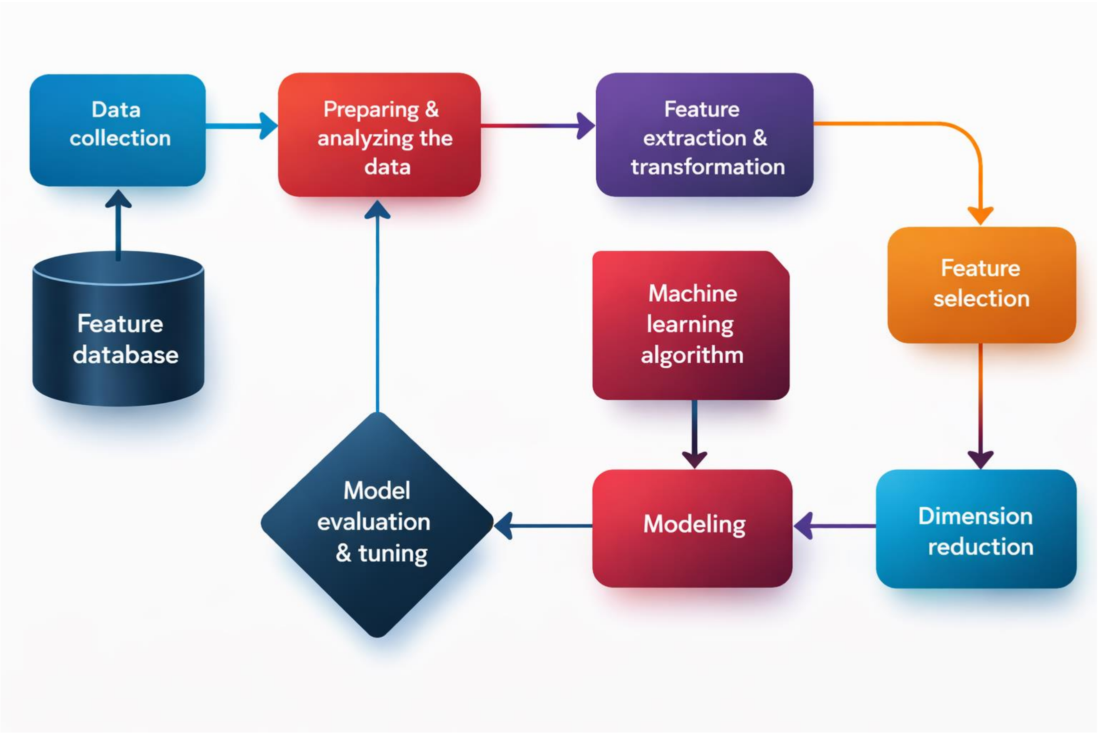
Why do we need proper metrics?

We need FSTs for every stage of the data evaluation process!



Why do we need proper metrics?

We need FSTs for every stage of the data evaluation process!



How can we transform the data?

Outline

0. Motivation
1. Digesting Measures: How does MLA decide?
2. **Meaningful measures: Truth beyond stats textbooks**
3. Recipes: Preparing real-world data for ML
4. Hands-on exercises

Recap

➔ Providing **meaningful data reports** is **crucial for creating confidence** in your scientific outcome!

➔ The majority of **ML-Approaches are making decisions** based on
1. distances (e.g. confidence intervals, k-means)
2. angles (correlation, covariances, regression, gradients, etc.)

➔ Data spaces should have **Euclidean properties** (orthonormal base)

➔ **Symmetry or normality** is often an **undeniable assumption**!

➔ **Disobeyance** of mathematical, physical, or statistical requirements may **obscure our outcome**!

PS: We will neither discuss exceptions (e.g., entropy-based ML) nor cover all needs for compositional data!

Let us concentrate on basic recipes for reliable ML applications!

Our playground: Data Types & Measures

Scale	Measure	Examples	Similarity measure	Central tendancy	Dispersion measure
Binomial	success or failure	Rainy days	incomparable, equality	$n \cdot p$	$np(1-p)$
(Multi-) nominal	memberships of categories	Soil types	membership, equality	mode, $\bar{x}_{geo}$, $n \cdot p_i$	entropy
Ordinal	as nominal plus < "is greater"	hurricane category	membership, rank order	$median\ Q_{50}$	$IQR = Q_{75} - Q_{25}$
Interval	distances $d(x_1,x_2)=\|x_1-x_2\|$	time, location	distance close to 0 , additive	*arithmetic mean:* $\bar{x}_{ar}$	*arithmetic stand. dev.:* s_{ar}
Rational	ratio: $f = {}^{x_1}/_{x_2}$	length, weight, isotope ratio	ratio close to 1, multiplicative	*geometric mean:* $\bar{x}_{geo}$	*geometric stand. dev.:* s_{geo}
Logistic	relative position between two limits: $\left(\frac{x-l}{u-x}\right)$	elevation, depth	log-ratio-distance, multiplicative	logistic mean $\bar{x}_{lgt}$	logistic variance s^2_{lgt}
Concentration	part of a constant sum	TOC, recovery rate	Aitchison distance, multiplicative	*geometric mean:* $\bar{x}_{geo}$	compositional variance

World of Σ

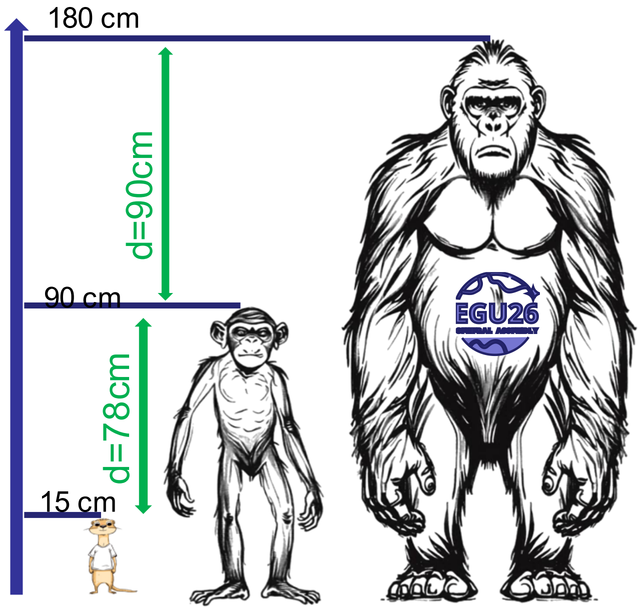
(Greek letter “sigma” = S
➔ sum of discrete numbers/objects)

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + \ldots + x_n$$

Common application:
- a and b are the sizes of two object.
- How large is their difference d?

Imagination:
- distance (in amount, length, distance, etc.)

Interpretation
-> the closer the two value, the more similar they are.



World of Σ

Native inhabitants:

signed integer $\boldsymbol{x} \, \epsilon \, \mathbb{Z} = \{a, -a : a \epsilon \mathbb{N}_0 \, \wedge \, a + (-a) = 0\}$ for discrete scales

rational number $\boldsymbol{x} \, \epsilon \, \mathbb{Q} = \{\frac{a}{b}, a, b \epsilon \mathbb{Z} \, \wedge \, b \neq 0\}$ for continuous scales

real number $\boldsymbol{x} \, \epsilon \, \mathbb{R}$: as rational numbers plus irrational numbers

Properties:

- **0** is the **center** (origin) and **neutral element** (*a+0=a* resp. *a-a =0*)
- **Distances** $|a - b| > 0, a, b \, \epsilon \, \mathbb{Q}/\mathbb{R}$ resp.
- $\| \vec{a} - \vec{b} \| > 0, : \vec{a}, \vec{b} \, \epsilon \, \mathbb{Q}^n/\mathbb{R}^n$ are **meaningful measures of similarity**

Operations +, -, * , / can be applied to all inhabitants without leaving the range of the scale!

➔Σ is symmetric: $|a - b| = |b - a|$

$\min(\Sigma) = -\infty \, \wedge \, \max(\Sigma) = \infty$: Home of Normality $\mathcal{N}(\mu, \sigma)$

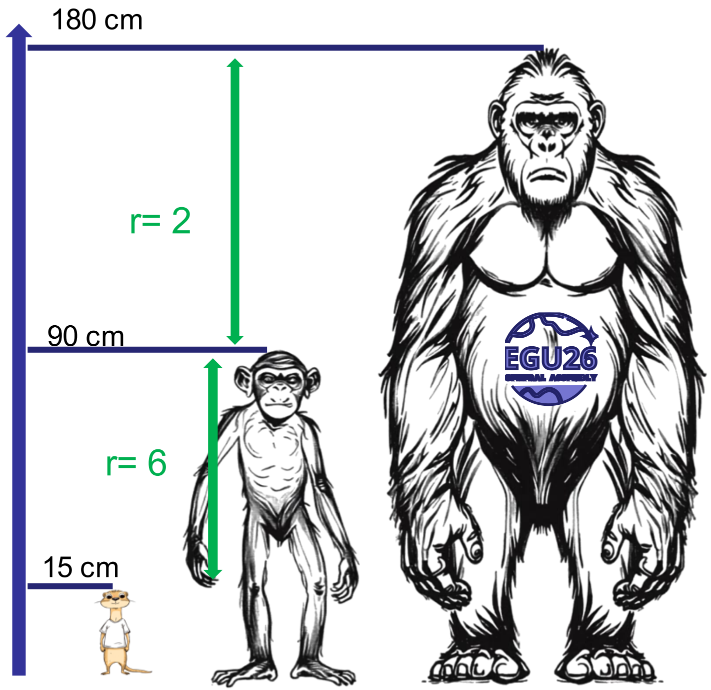
World of Π

(Greek letter "pi" = p
➔ product of discrete numbers/objects)

Common application:
a and b are the sizes of two object.
What is their relation (factor, ratio) r?

Thus, we calculate: $a * r = b$ or $b : a = b * a^{-1} = r$.
➔ two complementary operators * and : (÷)

Interpretation
➔ the closer the ratio to 1, the more similar are objects!



World of Π

$$\prod_{i=1}^{n} x_i = x_1 \cdot x_2 \cdot \dots \cdot x_n$$

Native inhabitants:
integer $\boldsymbol{x} \,\epsilon\, \mathbb{N} = \{1,2,3,\dots\}$ for discrete scales
positive rational number $\boldsymbol{x} \,\epsilon\, \mathbb{Q}_+ = \{\frac{\mathrm{a}}{\mathrm{b}} : \mathrm{a}, \mathrm{b} \neq 0\}$ for continuous scales
positive real number $\boldsymbol{x} \,\epsilon\, \mathbb{R}_+$

Properties:
- **1** is the **center** (origin) and **neutral element** (*a*1=a* resp. *a* * a^{-1} = *1*)
- **Ratios** $\frac{\mathrm{a}}{\mathrm{b}} > 0,\ \mathrm{a}, \mathrm{b} \,\epsilon\, \mathbb{Q}_+ \vee \mathbb{R}_+$ **are meaningful measures of similarity**

only multiplication /division */ can be applied to all inhabitants without leaving the range of the scale!

Π **is asymetric**: $\forall\, a > 1 \;\; \exists\, \mathrm{a}^{-1} \,\epsilon\,]0,1[_{\mathbb{R}}$

$$\Rightarrow |\mathrm{a}| \neq |\mathrm{a}^{-1}| = \left|\frac{1}{\mathrm{a}}\right|; \qquad \text{e.g. } \left|\frac{1}{2}\right| \neq \left|\frac{2}{1}\right|$$

Arithmetic mean is tied to full real space!

Let's look at the arithmetic mean:

$$\bar{x}_a = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$n \cdot \bar{x}_a = \sum_{i=1}^{n} x_i$$

$$0 = \left(\sum_{i=1}^{n} x_i\right) - (\bar{x}_a + \bar{x}_a + \cdots + \bar{x}_a)$$

$$0 = \sum_{i=1}^{n} (x_i - \bar{x}_a)$$

$$\sum_{i:x_i<\bar{\bar{x}}} (x_i - \bar{x}_a) = -\sum_{i:x_i>\bar{\bar{x}}} (x_i - \bar{x}_a)$$

➔ **The arithmetic mean is meaningful in a symmetrical feature space, only!**

methods – vs. data – space: Abilities and Disabilities

$\Sigma : \mathbb{R}, \mathbb{Q}(\mathbb{Z})$

Sums and differences may result in **negative amounts**!
Thus, Σ calls for **full real/rational space!**

The **most popular representatives** are the **arithmetic mean** and the arithmetic **standard deviation**.

➔They **pass over** their properties and limitations to the **absolute majority of statistical methods**!

Σ is the **home for normality** which requires **free vision to** $\pm\infty$

Predominant home for **statistical methods** and **Textbooks** about Stats!

$\Pi : \mathbb{R}^+, \mathbb{Q}^+(, \mathbb{N})$

Positive numbers cannot leave the **positive branch** by multiplication/division!

Π has only a few representatives but cares for the **absolute majority of real-world data**!

The implicit **skewness hamper linearity!**

Π is the **home for log-normality**

Predominant home for **real-world data** and **empirical Sciences!**

Bridging the gap

Ways out of the mess!

$$\boldsymbol{\Sigma} : \mathbb{R}, \mathbb{Q}$$

Home of **methods**

$$\Pi : \mathbb{R}^{+}, \mathbb{Q}^{+}, \mathbb{N}$$

Home of real-world **data**

Bridging the gap

Let us start from real world multiplicative (rational) feature space:

a: **1** is the **center** and the **neutral element** of the rational scale!
b: **the closer** the ratio to 1, **the more** similar the objects are!

but:
➔ Π is asymmetric:

for two measure $0 < a < b$: $$\left|\frac{a}{b}\right| < \left|\frac{b}{a}\right|$$

but: $$\left|\log\left(\frac{a}{b}\right)\right| = \left|\log\left(\frac{b}{a}\right)\right|$$

because: $$\left|\log\left(\frac{a}{b}\right)\right| = |\log a - \log b| = |\log b - \log a| = \left|\log\left(\frac{b}{a}\right)\right|$$

The log-ratio of two measures is a symmetric measure!

The ratio is meaningful, but bends the feature space!

Bridging the gap

The log-ratio of two measures is a symmetric measure!

Therefore, the **arithmetic mean of log-transformed rational data** should be meaningful as well:

$$x \rightarrow x' = \log(x): \qquad \bar{x}'_{ar} = \frac{1}{n}\sum \log x$$

But, a log-mean is **difficult to imagine**. Let us transform the log-mean back:

$$x' \rightarrow x = e^{x'}: \qquad e^{\bar{x}'_{ar}} = e^{\frac{1}{n}\sum \log x_i} = \left(\prod_{i=1}^{n} e^{\log x_i}\right)^{\frac{1}{n}} = \sqrt[n]{x_1 \cdot x_2 \ldots x_n} = \boldsymbol{\bar{x}_{geo}}$$

The geometric mean is meaningful on multiplicative (ratio) feature scales!

Ways out of the mess!

$$\boldsymbol{\Sigma} : \mathbb{R}, \mathbb{Q}$$

Home of **methods**

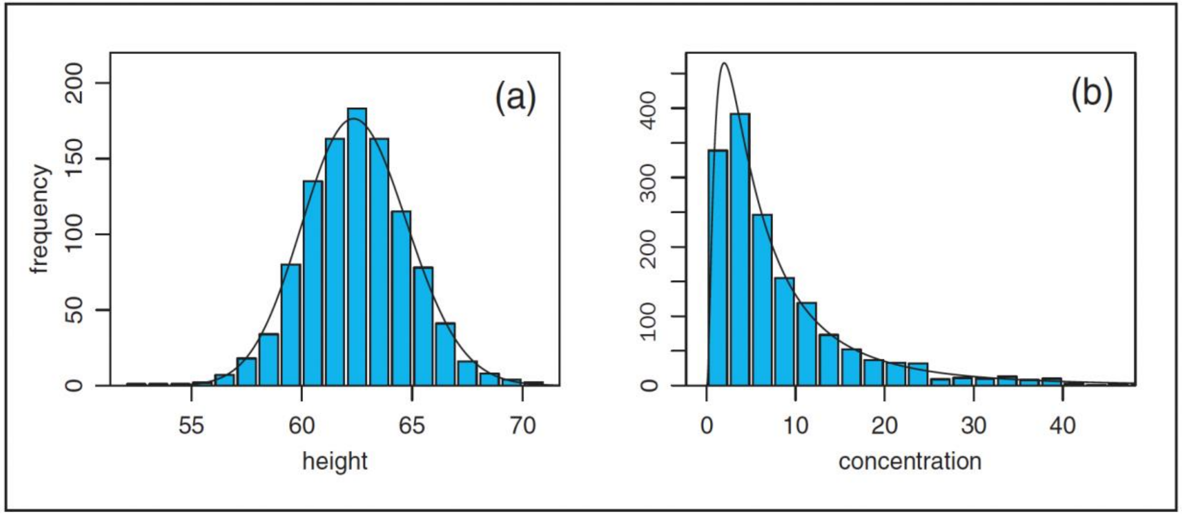
$$\Pi : \mathbb{R}^+, \mathbb{Q}^+, \mathbb{N}$$

Home of real-world **data**

log(x)

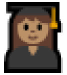
e^x

Normal or log-Normal?



Distribution fits the Normal with p = 0.75 and the log-Normal with p = 0.74

Distribution fits the log-Normal with p = 0.41 but not the Normal (p <0.0001)



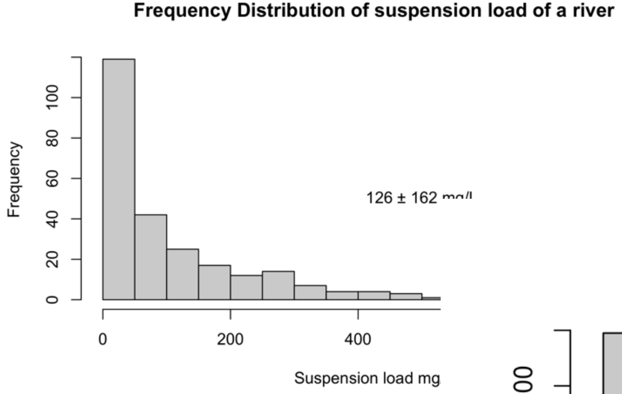
Takeaway:

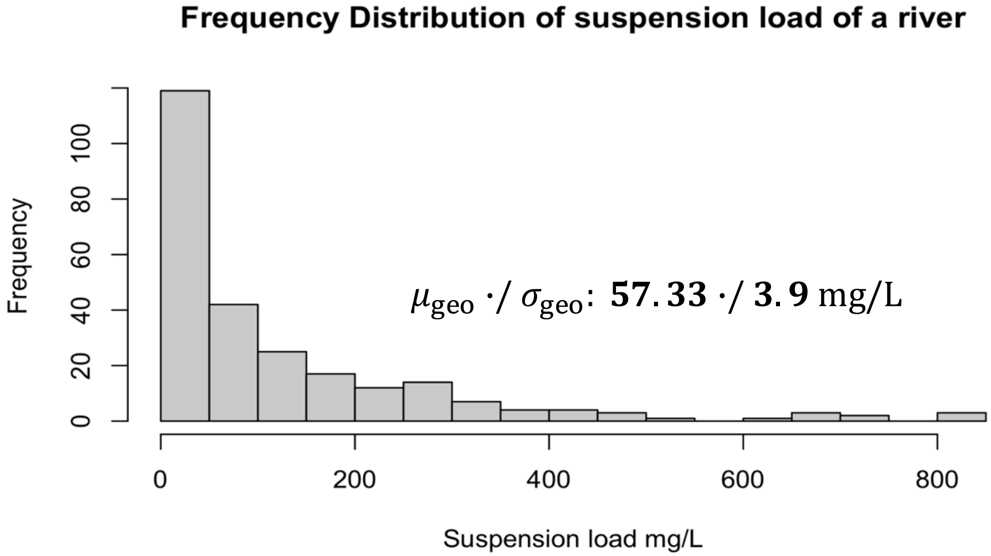
- **Assuming the log-Normal doesn't hurt, but the Normal!**
- **You can infinitely flatten a curve, but never bend a straight line!**

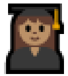
Limpert, E., Stahel, W.A., Abbt, M., 2001. Log-normal Distributions across the Sciences: Keys and Clues. BioScience 51, 341. https://doi.org/10.1641/0006-3568(2001)051%255B0341:LNDATS%255D2.0.CO;2

1-σ for geometric mean

Recipe:
1. transform your data: log(x)
2. analyse yor data
3. transform desired parameters back: e^x







But, keep in mind:
$\mathbb{R}^+$ is multiplicative

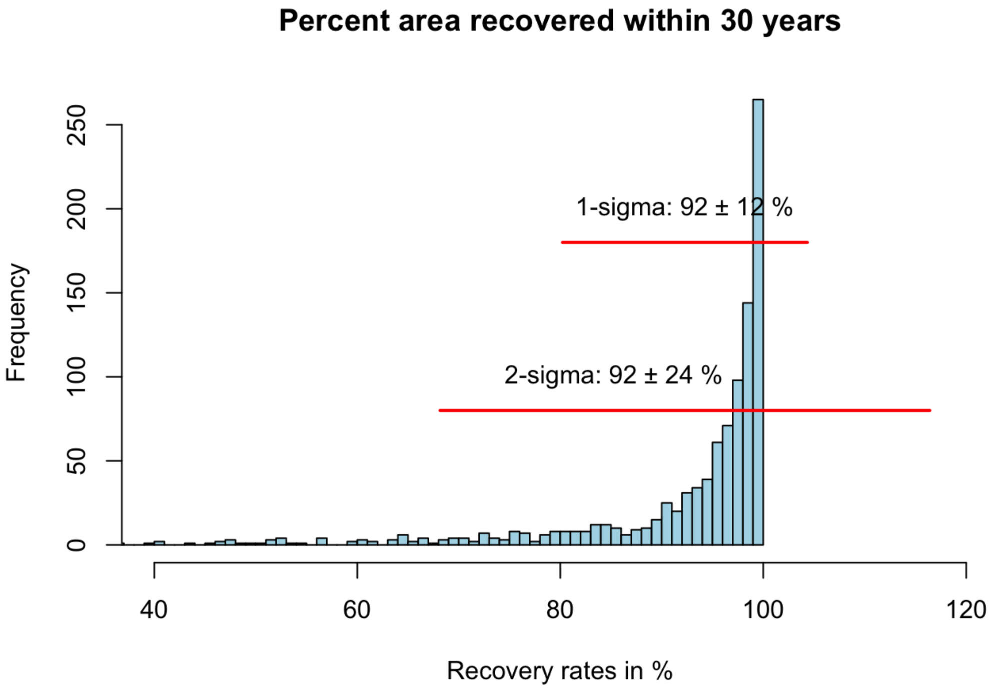
Property	Normal distribution (Gaussian, or additive normal, distribution)	Log-normal distribution (Multiplicative normal distribution)
Effects (central limit theorem)	Additive	Multiplicative
Shape of distribution	Symmetrical	Skewed
Models		
Triangle shape	Isosceles	Scalene
Effects at each decision point x	$x \pm c$	$x\ {}^{\times}\!/\ c'$
Characterization		
Mean	$\bar{x}$, Arithmetic	$\bar{x}^*$, Geometric
Standard deviation	s, Additive	s*, Multiplicative
Measure of dispersion	$cv = s/\bar{x}$	s*
Interval of confidence		
68.3%	$\bar{x} \pm s$	$\bar{x}^*\ {}^{\times}\!/\ s^*$
95.5%	$\bar{x} \pm 2s$	$\bar{x}^*\ {}^{\times}\!/\ (s^*)^2$
99.7%	$\bar{x} \pm 3s$	$\bar{x}^*\ {}^{\times}\!/\ (s^*)^3$

Notes: cv = coefficient of variation; ${}^{\times}\!/$ = times/divide, corresponding to plus/minus for the established sign ±.

Is this brand-new science? Definitly not:
Mean, Normal-Distribution and CLT published by C-F Gauss (**1809**) and Geometric Mean, Sd and CLT for Log-Normal D. McAllister **1879**;
The Relation between both has been reported by F. Galton **1989**

Limpert, E., Stahel, W.A., Abbt, M., 2001. Log-normal Distributions across the Sciences: Keys and Clues. BioScience 51, 341.
https://doi.org/10.1641/0006-3568(2001)051%255B0341:LNDATS%255D2.0.CO;2

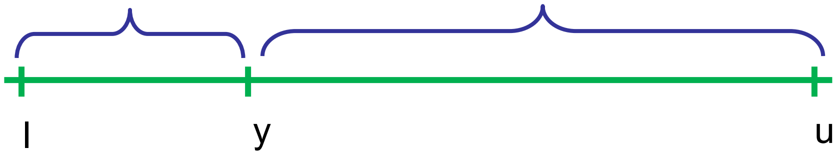
Double constraint data: the logistic moments



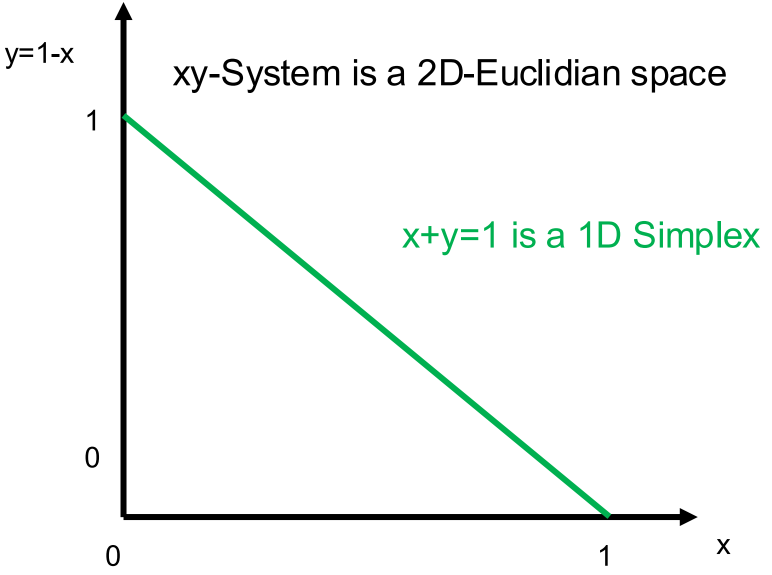
Here we have a lower and an upper limit, which can not be exceeded in a meaningful way!

→every value y has two information: distance to the left and to the **right boundary**!

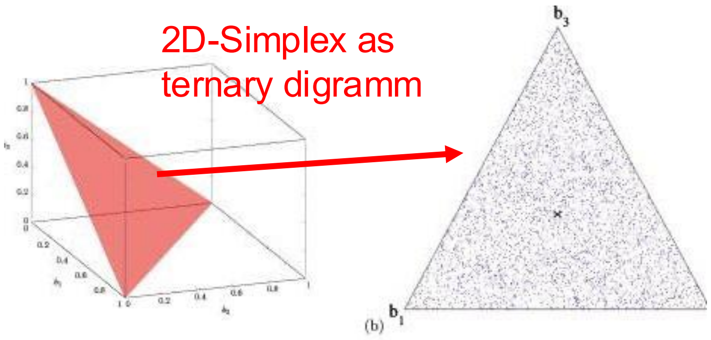
Both informations are not independent!

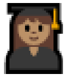


Double constraint Spaces are compositional by nature



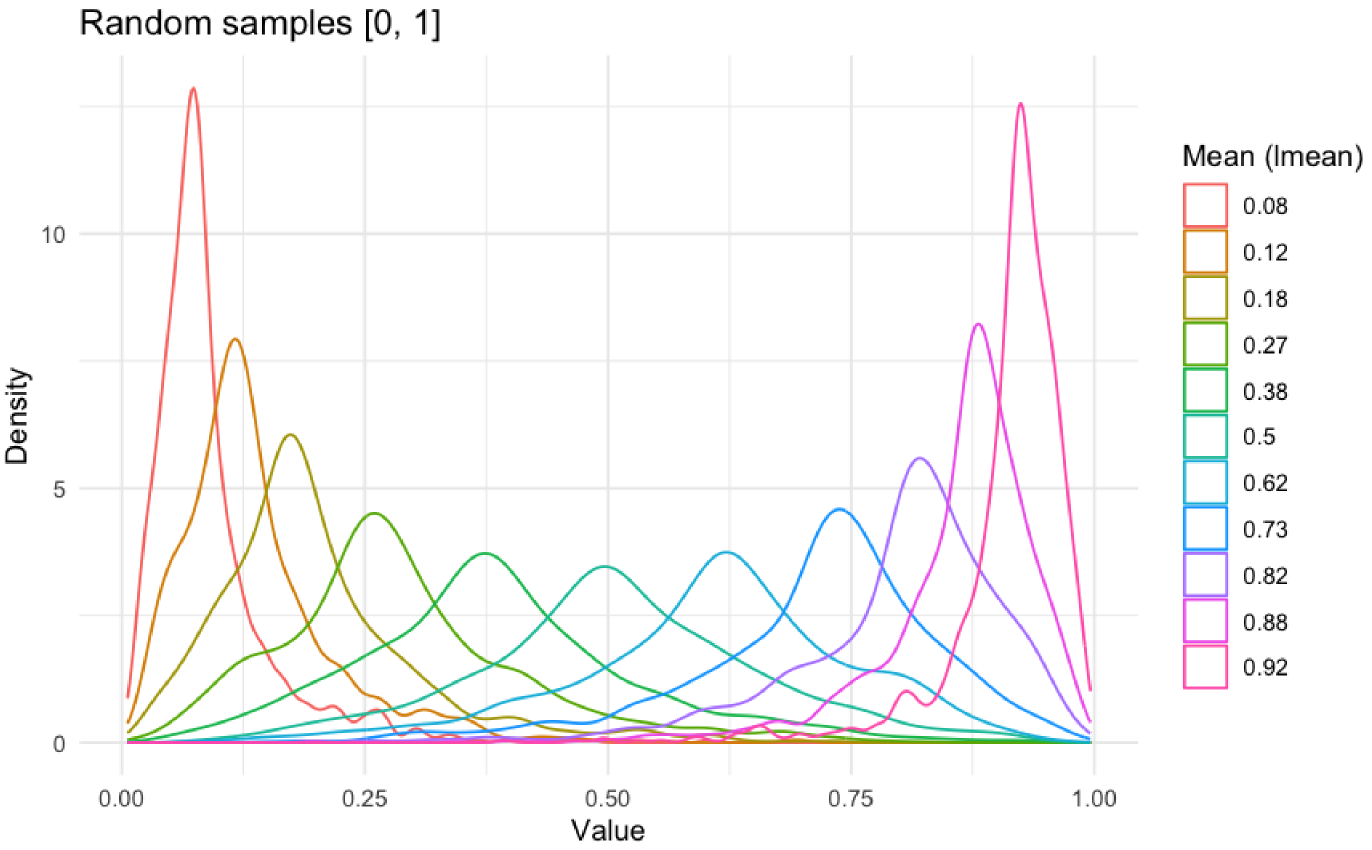
3D-Space with x+y+z=1

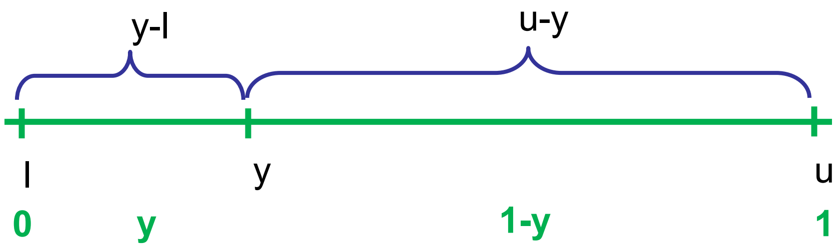




Feature spaces have an upper limit!
➔They are **constraint by a constant sum**!

Useful solution: the logistic normal





But we can open our double-bounded feature scale by using a logratio transformation:

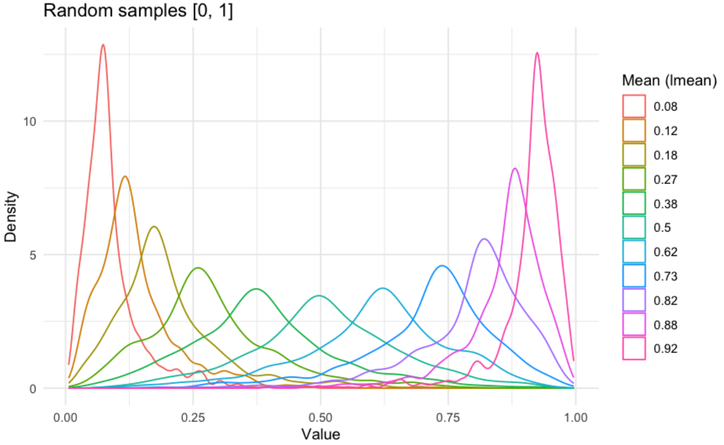
$$y' = \log\left(\frac{y - l}{u - y}\right) \qquad y' = \log\left(\frac{y}{1 - y}\right) : \; y \in]0,1[$$

Calculate your parameter p (= mean, variance, σ-intervals) and transform derived values back:

$$y = l + \frac{u \cdot e^p}{1 + e^p} \qquad y = \frac{e^p}{1+e^p} : y \in]0,1[$$

Johnson, N.L., 1949. Systems of Frequency Curves Generated by Methods of Translation. Biometrika 36, 149–176.

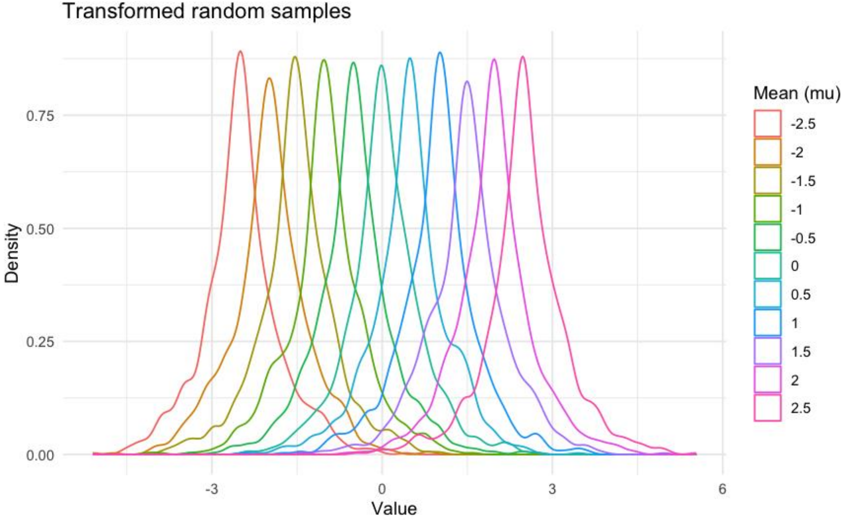
The logistic moments



$$y \to y' \epsilon \mathbb{R}$$

$$y' = \log\left(\frac{y}{1-y}\right)$$

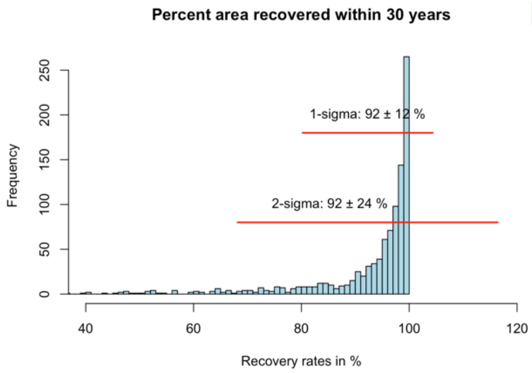
full properties of an interval scale



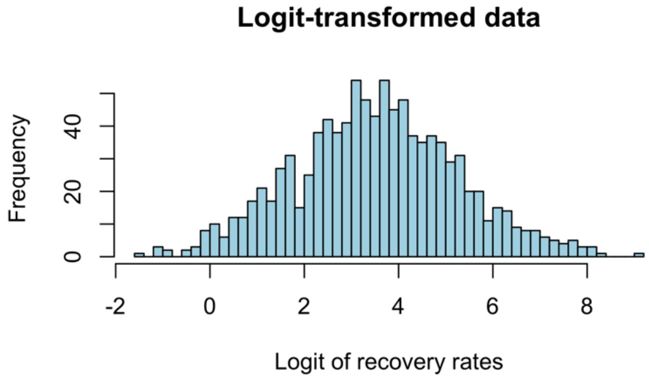
$$y = \frac{e^{y'}}{1 + e^{y'}}$$

$$y' \to y \, \epsilon \,]0.1[_{\mathbb{R}_+}$$

Meaningful intervals:

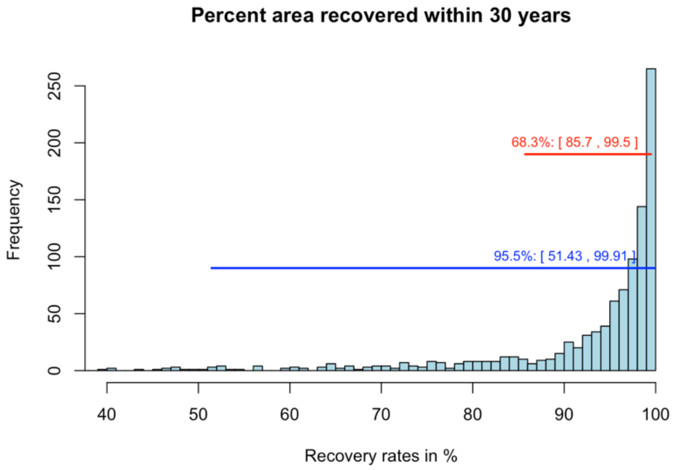


$$y' = \log\left(\frac{y}{1-y}\right)$$



$$[\mu \pm \sigma]$$

$$y = \frac{e^{y'}}{1 + e^{y'}}$$



Take aways (FST)

- **Most real-world data** is only **meaningful in** $\mathbb{R}^+$
- $\mathbb{R}^+$ indicates **multiplicative similarity** measure
- $\mathbb{R}^+$ indicates **skewed distance** measure
- Most **statistical methods** assuming data living in full $\mathbb{R}$
- $\mathbb{R}$ provides **additive, symmetric distances** as **similarity measures**
- **log()** transforms $\mathbb{R}^+ \rightarrow \mathbb{R}$ and turns **multiplication into addition**
- **exp()** transforms $\mathbb{R} \rightarrow \mathbb{R}^+$ and turns **addition into multiplication**
- **disobeying mathematical rules** results in **senseless** informations
- a high amount of **real-world data** has an **upper limit u** in $\mathbb{R}^+$
- double constraint data (**DCD**) is naturally 2-dimensional (or 2n-D)
- **DCD** is multiplicative (ratios are meaningful)
- **log-ratio-** (logistic-) transformation **opens DCD to** $\mathbb{R}$
- Ignoring FS-constraints leads to **meaningless results**

Requirements of feature scales for correlation and linear regression:

- **Regressend** or **dependent variable** has to be **normal!**
- Predictor variables have to be at least **symmetric distributed**, ideally normal!
- all scales have to be **interval scales** resp. **additive unlimited!**
- arithmetic **mean** and **standard deviation** have to be **meaningful!**

Sounds challenging with real world data!

In particular, **double-constraint feature scales have a compositional nature**, indicating **spurious correlation** under the constant-sum constraint.

Correlation, Cosine, Scalar product

let us consider $\vec{x} = ((x_1 - \bar{x}), (x_2 - \bar{x}), \cdots, (x_n - \bar{x}))$ and
$\vec{y} = ((y_1 - \bar{y}), (y_2 - \bar{y}), \cdots, (y_n - \bar{y}))$ as 2 vectors $\mathbb{R}^n$

<u>Then we may interpret the correlation coefficient as cosine of the angle between $\vec{x}$ and $\vec{y}$:</u>

$$\cos\sphericalangle(\vec{x}, \vec{y}) = \frac{<\vec{x}, \vec{y}>}{\|\vec{x}\|\|\vec{y}\|} = \frac{\sum_{i=1}^{n} x_i y_i}{\sqrt{\sum_{i=1}^{n} x_i^2}\sqrt{\sum_{i=1}^{n} y_i^2}} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})_i^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})_i^2}} = r$$

$<\vec{x}, \vec{y}>$: Scalar product and $\| \vec{x} \|$ is the norm a the vector. .

Extreme cases:

1. case: r=0 ➔ $\cos\sphericalangle(\vec{x}, \vec{y})=0$ ➔ $\sphericalangle(\vec{x}, \vec{y}) = 90°$
   ➔ both variable vectors are rectangular to each other!
2. case: r=1 ➔ $\cos\sphericalangle(\vec{x}, \vec{y})=1$ ➔ $\sphericalangle(\vec{x}, \vec{y}) = 0°$
   ➔ borth variable vectors point to the same direction
3. Fall: r=-1 ➔ $\cos\sphericalangle(\vec{x}, \vec{y})=-1$ ➔ $\sphericalangle(\vec{x}, \vec{y}) = 180°$
   ➔ both variable vectors point to the same direction

Our data space require Euclidian properties: <u>an orthonormal base</u>!
(spurious correlations contradict these requirements!

Correlation, Cosine, Scalar product

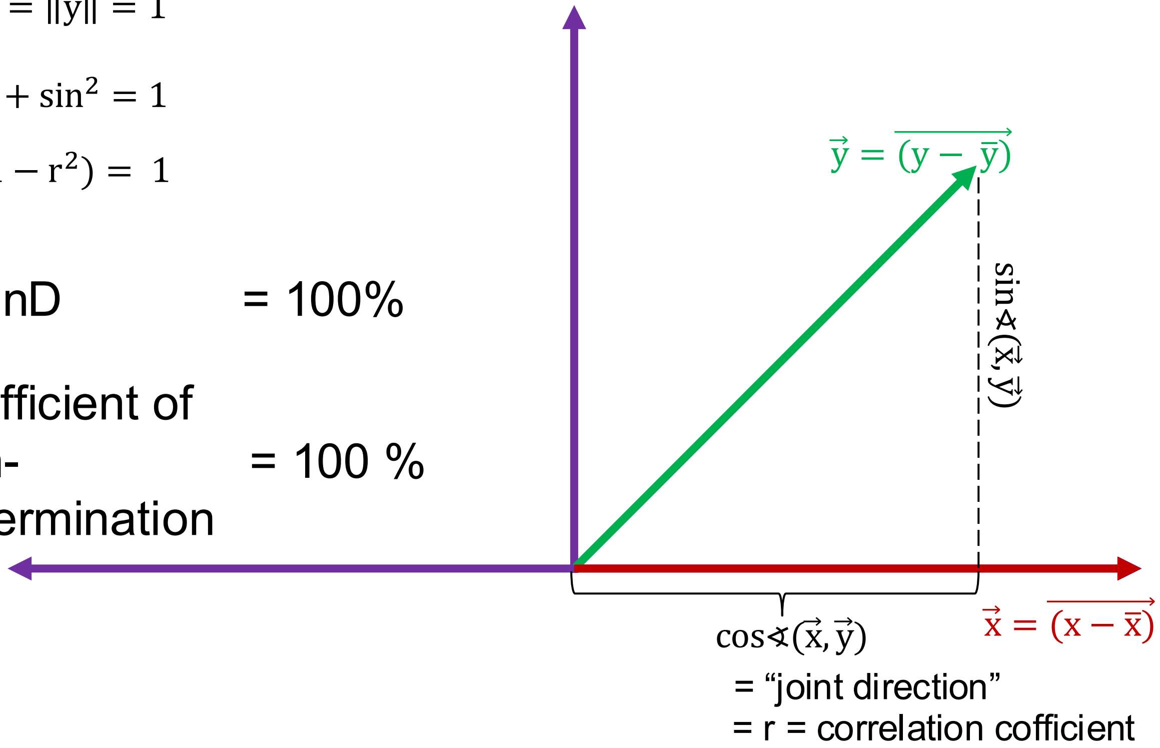
applying Pythagoras for $\|\vec{x}\| = \|\vec{y}\| = 1$

$$\cos^2 + \sin^2 = 1$$

$$r^2 + (1 - r^2) = 1$$

COD + COnD = 100%

coefficient of determination" (R^2) + coefficient of non-determination = 100 %

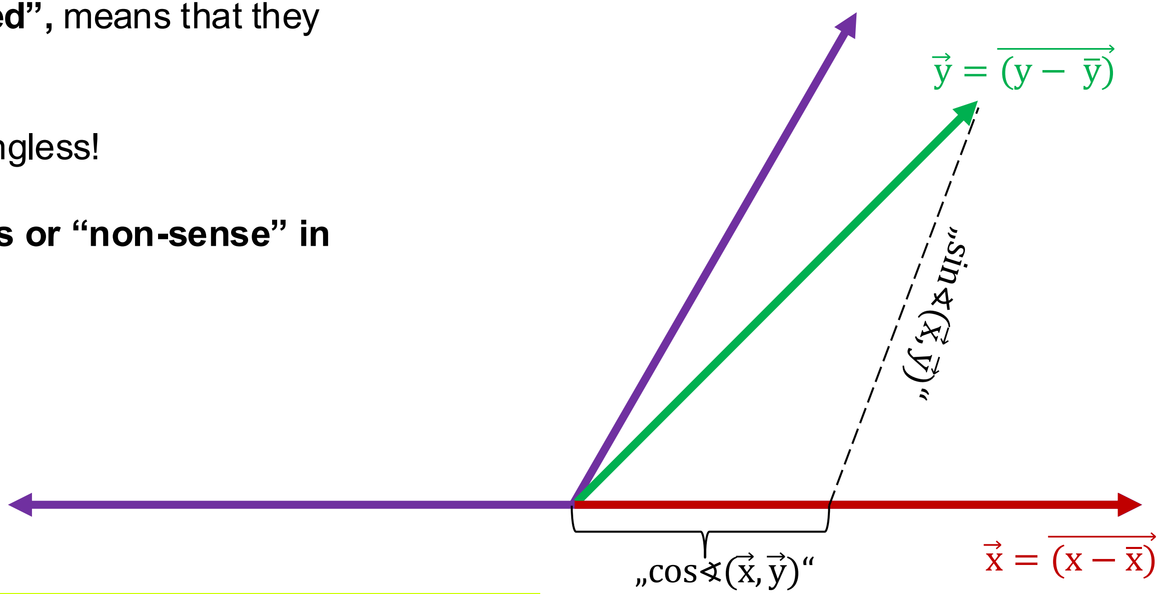


Correlation, Cosine, Scalar product

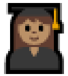
If two variables are **“spurious correlated”,** means that they bases are **not rectangular**!

➔ cos, correlation, and COD are meaningless!

➔ **Any further correlation is senseless or “non-sense” in sensu strictu!**



Spurious correlations obscure the message in our data!



Spurious correlations produce artificial pattern!

Logratio-transformations are able to produce orthonormal bases (Euclidian metrics)

Outline

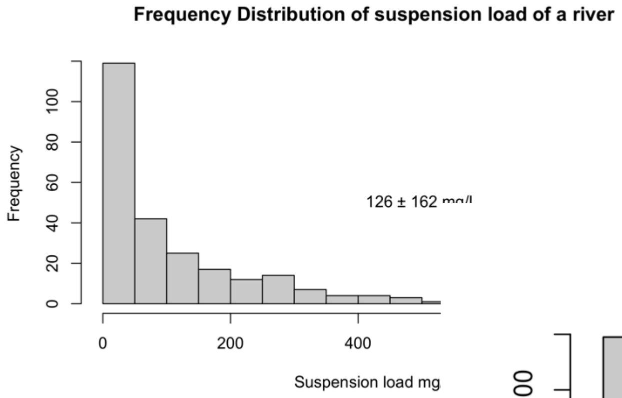
0. Motivation
1. Digesting Measures: How does MLA decide?
2. Meaningful measures: Truth beyond stats textbooks
3. Recipes: Preparing real-world data for ML
4. Hands-on exercises

Recipe 1: 0 < x

- **Step 1**: <u>**Check the distribution**</u> of the data using histograms, boxplots, or Q-Q plots to assess skewness and identify potential outliers. If the data has a lower limit greater than zero, shift the data by adding a constant to all values to ensure that all values are positive before applying the log-transformation.
- **Step 2**: Apply the <u>**log-transformation**</u> to the data using the formula $\boldsymbol{x'}$**=log(**$\boldsymbol{x}$**)**, where x is the original data and x' is the transformed data.
- **Step 3**: After log-transformation, <u>**check the distribution of the transformed data**</u> to ensure that it is more symmetric and closer to a normal distribution.
- **Step 4**: <u>**Calculate the arithmetic mean and standard deviation**</u> of the log-transformed data, which are now appropriate measures of central tendency and variability for the transformed data.
- **Step 5: <u>Back-transform the results</u>** to the original scale by applying the **exponential function** $\boldsymbol{x = e^p}$ to the mean $(\mathrm{p} = \bar{\mathrm{x}}_{\log(\mathrm{x})})$ and standard deviation $(\mathrm{p} = \mathrm{s}_{\log(\mathrm{x})})$of the log-transformed data.
- After back transformation, the mean will be the **geometric mean** $\boldsymbol{\bar{x}_{geo}}$of the original data, and the **standard deviation** $\boldsymbol{s_{geo}}$ will be the geometric standard deviation of the original data.
- Due to the **direct multiplicative nature** of your original data, you can now report the 1-sigma range of the original data as the geometric mean */ geometric standard deviation: $\boldsymbol{\bar{x}_{geo}} */ \boldsymbol{s_{geo}}$.

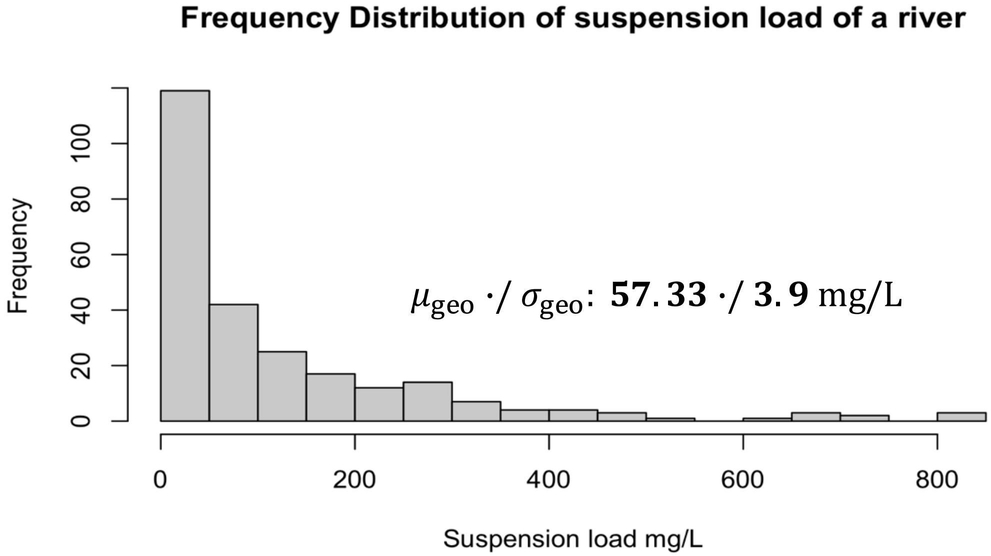
➔ <u>Even applying higher sigma intervals or small sample sizes, these ranges cannot exceed ℝ+ resp. ℚ+</u>

Recipe 1: $0 < x$



Recipe:
1. transform your data: log(x)
2. analyse yor data
3. transform desired parameters back: e^x

But, keep in mind:
$\mathbb{R}^+$ **is multiplicative**



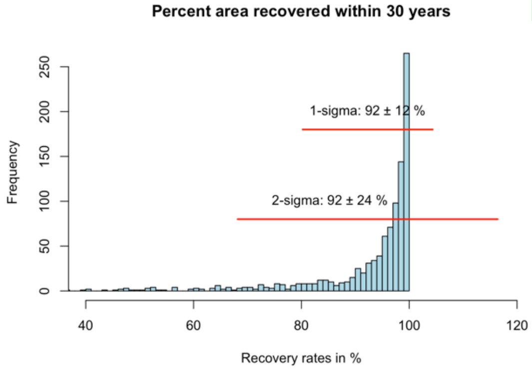
Recipe 2: lower limit < x < upper limit

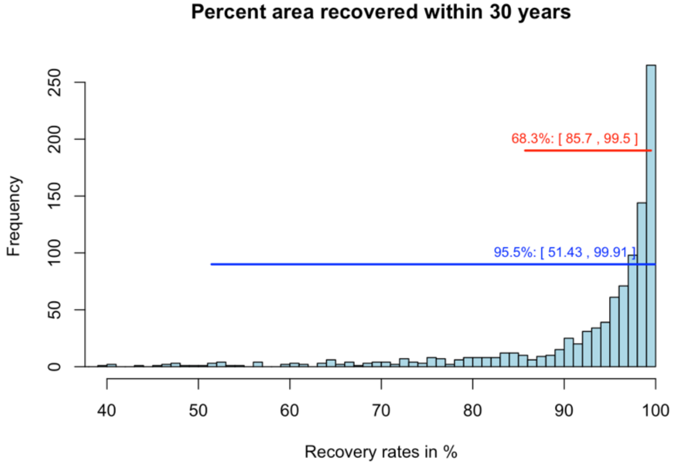
- **Step 1: Check the distribution** of the data using histograms, boxplots, or Q-Q plots to assess skewness and identify potential outliers. If the data are proportions, ensure that all values are between 0 and 1. If there are values of 0 or 1, consider applying a small adjustment (e.g., adding a small constant to 0 values and subtracting a small constant from 1 values) to avoid issues with the logit transformation.
- **Step 2a: (optional) If your data has a lower and/or upper limit** that is **not 0 or 1,** perform a min-max normalization to scale the data to the]0,1[interval before applying the logit transformation. Therefore you have to apply the following transformation to your data:

$$x' = x - \min(x) / \max(x) - \min(x).$$

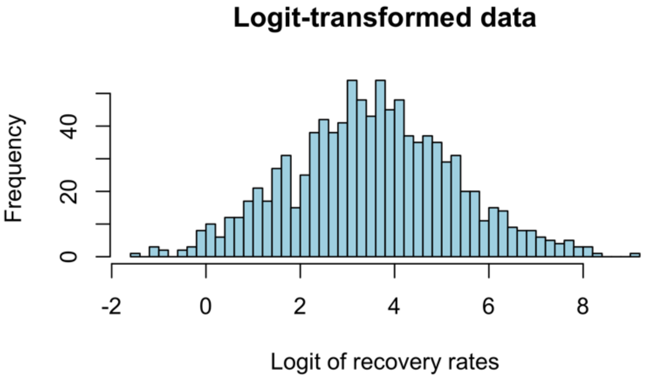
- **Step 2: Apply the logit transformation** to the data using the formula $x' = \log(x/1 - x)$, where x is the original data and x' is the transformed data. Hereby, exclude the minimum and maximum values to avoid issues like zero and zero divisor.
- **Step 3:** After logit transformation, **check the distribution** of the transformed data to ensure that it is more symmetric and closer to a normal distribution. If there is still a considerable skewness, you may consider applying e.g. a square-root, an arctan() or an arcsin() transfomation data. If the data is symmetric but with an obvious smaller or larger variance, you may consider applying a scaling.
- **Step 4: Calculate the arithmetic mean and standard deviation on the logistic scale**, which are now appropriate measures. Do not interpret these results. Just calculate the 1-sigma-ranges on the logistic scale as mean ± standard deviation in the logistic FS.
- **Step 5: Back-transform the results to the original scale** by applying the inverse logit function to the mean and both bounds of the 1-sigma interval of the logit-transformed data.
- Due to the nature of the logit transformation, these back-transformed values will always be between 0 and 1, ensuring that they are meaningful for proportions and compositional data.
- **Step 5b: (optional) If you performed a min-max normalization in Step 2a,** back-transform the results to the original scale by applying the inverse of the min-max normalization.

Recipe 2: lower limit < x < upper limit





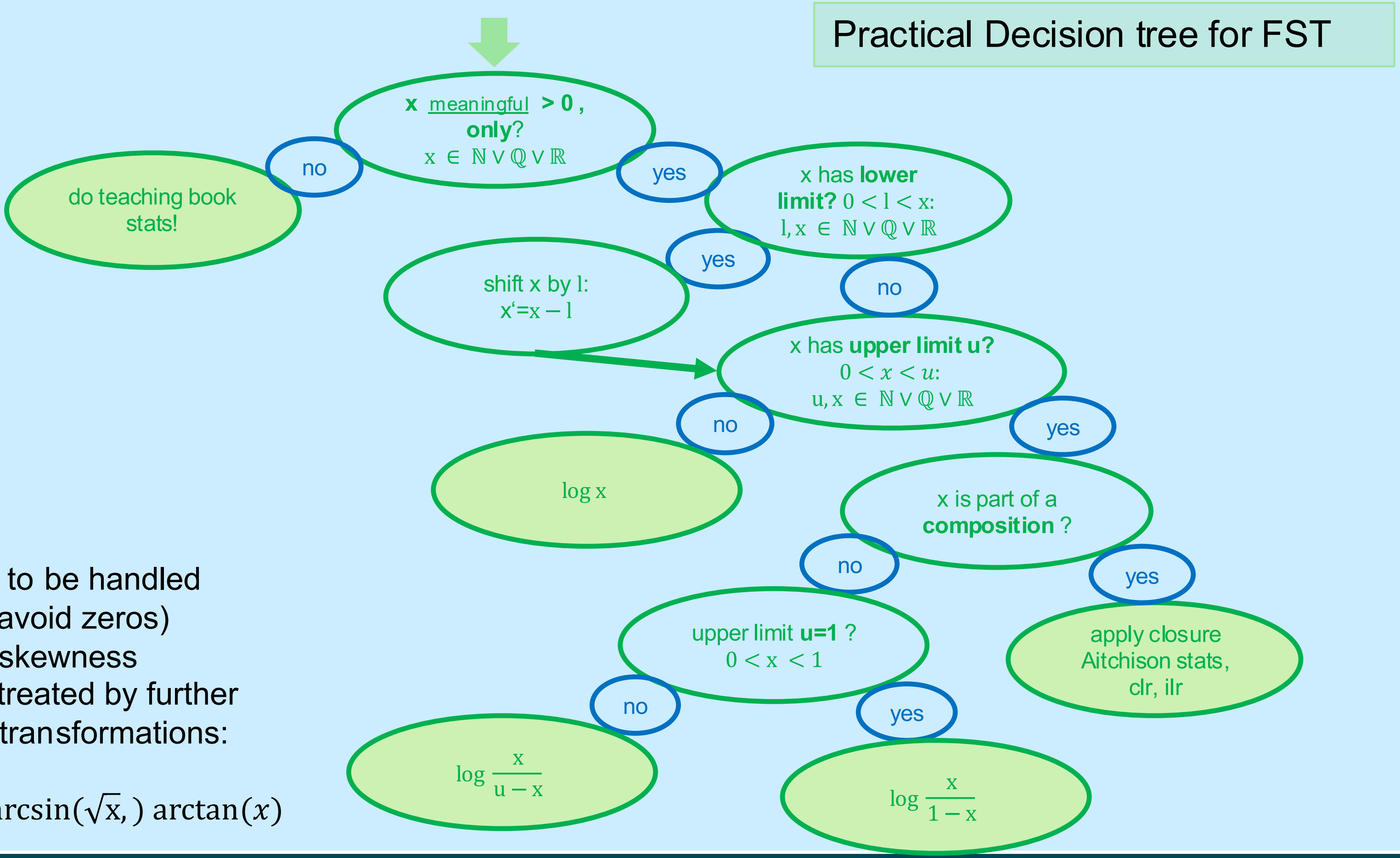
$$y' = \log\left(\frac{y}{1-y}\right)$$



$$[\mu \pm \sigma]$$

$$y = \frac{e^{y'}}{1 + e^{y'}}$$

Decision tree for FST



Remarks:

1. limits have to be handled with care (avoid zeros)
2. remaining skewness should be treated by further non-linear transformations: e.g. $\arcsin(x), \arcsin(\sqrt{x},)\ \arctan(x)$

Other useful transformations:

!! For shape control only: They are not capable of solving algebraic constraints !!

1. Square root: $y' = \sqrt{y}$ positive skewness and kurtosis, variable remains in $\mathbb{R}^+$

2. Reciprocal: $y' = \frac{1}{y}$ positive skewness and kurtosis, variable remains in either in $\mathbb{R}^+ \; or \; \mathbb{R}^-$

3. Arcsine & Arcsine-square root: $y' = \arcsin y$ & $y' = \frac{2}{\pi} \arcsin \sqrt{y}$ positive skewness in $]0{,}1[_{\mathbb{R}^+}$

4. Box-Cox-Transformation for $\frac{\mathrm{x}^{\lambda}-1}{\lambda}, \lambda \neq 0 \wedge \mathrm{x} > 0$

Only in combination with log and logistic-transformation:

5. upper boundary: $y' = max - y$. negative skewness in combination with e.g. log; $\mathrm{from}\ \mathbb{R}^+]0, u[\ to \mathbb{R}^+]0{,}1[$

6. Adding constant: $y' = y + a$. translation for e.g. avoiding 0-value for log or reciprocal

plus **standardization** and **normalization** technics but do not help for non-normality, skewness or algebraic constraints!!! (an overview is given e.g. Minkyungs'sblog)

cf. also Zhang et al, 2024

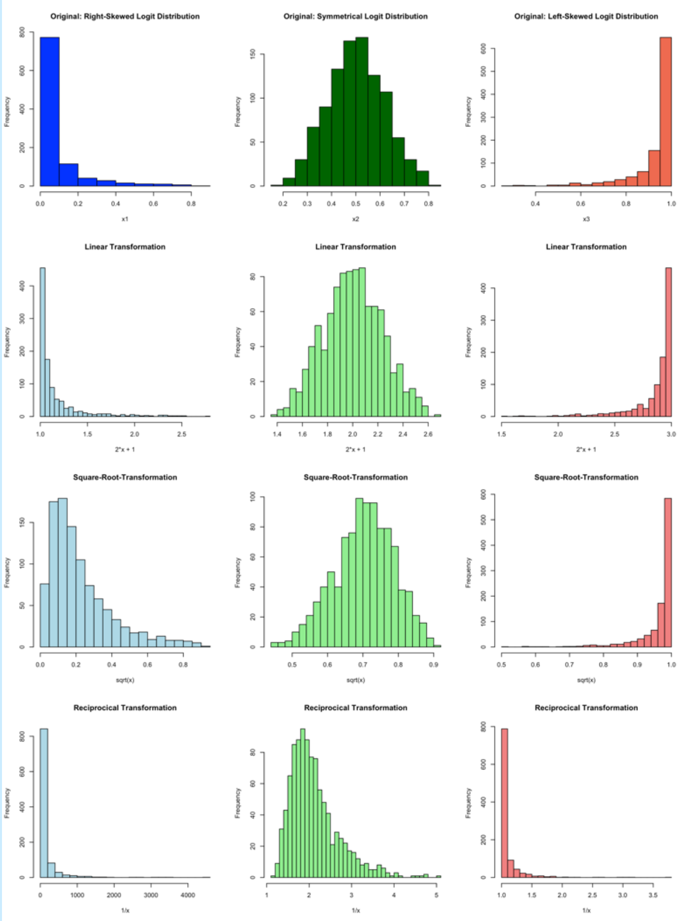
Additional transformations

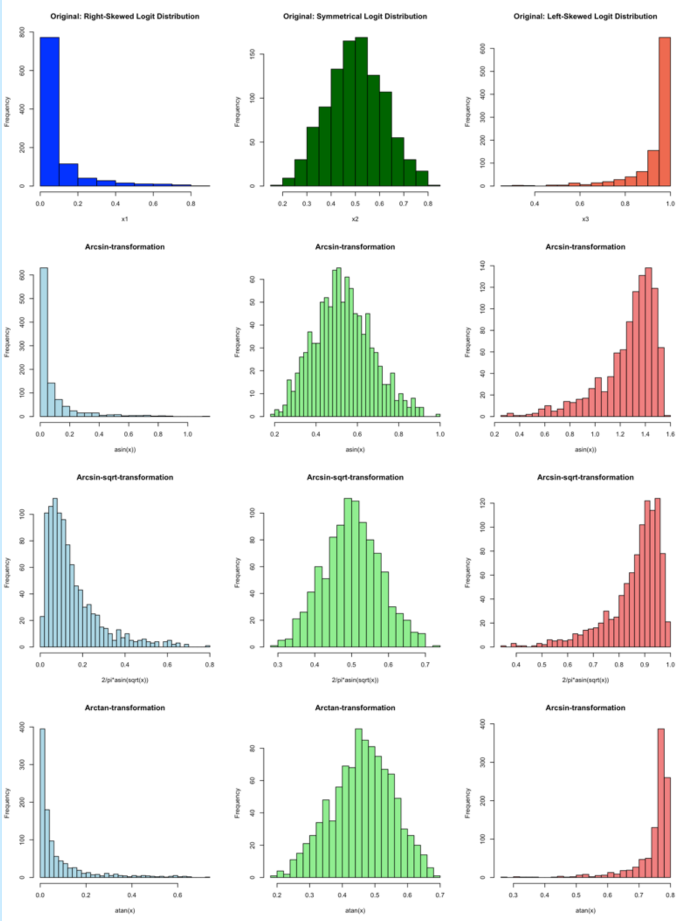
original

$x^2 + 1$

$\sqrt{x}$

$1/x$





original

asin x

$\frac{2}{\pi} \operatorname{asin}\sqrt{x}$

atan x

Additional transformations

original

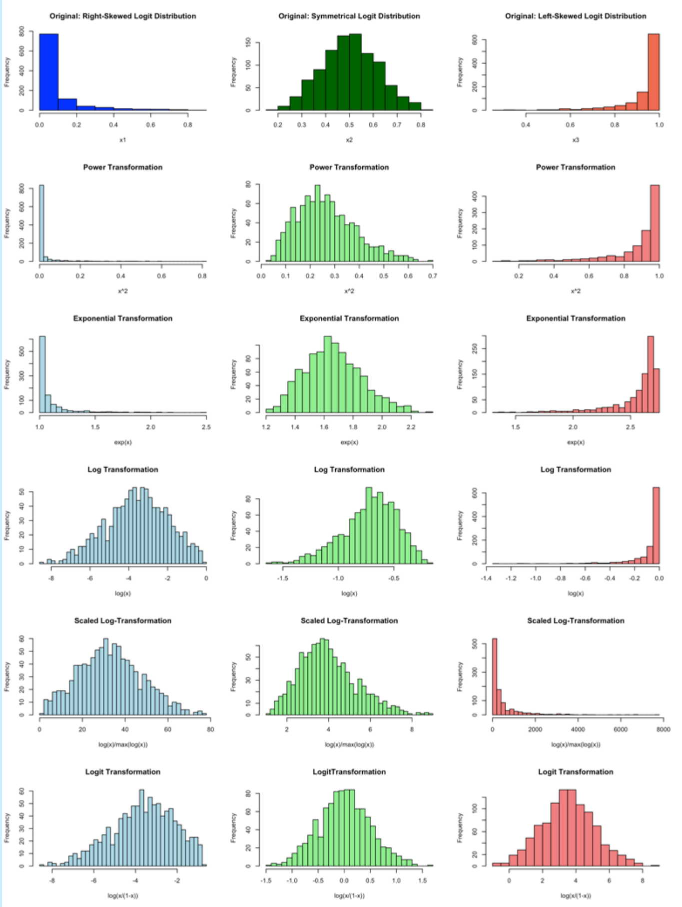
x^2

e^x

$\log x$

$$\frac{\log x}{\max(\log x)}$$

$$\frac{x}{1-x}$$



Here, **only the logit-transformation** preserves and produces **symmetry**!

Outline

0. Motivation
1. Digesting Measures: How does MLA decide?
2. Meaningful measures: Truth beyond stats textbooks
3. Recipes: Preparing real-world data for ML
4. Hands-on exercises

Hands-on exercise

Open the exercise file “exercise_EGU26_SC2-28”
as **R-Markdown** or **Jupyter-Notebook**
at our Git-hub repo

https://github.com/soga-lab/EGU2026_SC

Work and discuss in small groups of 4-5 people!

Written reflection and feedback

Thanks for your attention!

For further feedback or questions:
soga@zedat.fu-berlin.de

TECHNISCHE UNIVERSITÄT BERLIN

Stiftung Innovation in der Hochschullehre

EGU26 GENERAL ASSEMBLY

FREIE UNIVERSITÄT BERLIN

Join at

slido.com

#2859 508

Passcode:

egu26

Written reflection and feedback